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Green Industrial Policy and Sustainable Development in Emerging Markets: A Multi-Agent Decision-Support System with International-Law Source Ranking for Evidence-Grounded Cross-Border Green Product Requirements

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Abstract

Emerging-market exporters encounter global sustainability governance not as a treaty or a target but as a product rule: an energy performance threshold, a restricted-substance limit, and a recyclability declaration. Those rules are published, and publication is not the same as access. A firm with a regulatory affairs department can establish which version of a measure is in force, which products it covers and what evidence the destination market will accept; the small- and medium-sized enterprises that dominate emerging-market export bases usually cannot, so rules written to raise environmental standards can exclude the firms least equipped to read them. This paper asks whether emerging digital technologies can convert the transparency infrastructure of the trading system into sustainable business capability. We develop GRACE, a WTO-informed decision-support system that resolves regulatory versions and timelines before interpretation begins, ranks evidence by the authority of its source, declines to state any obligation that no official passage supports, and escalates to human experts when the record is incomplete. Evaluation covers 214 held-out notification families under family-level splits and five seeds, against direct prompting, generic retrieval-augmented generation, a domain-adapted single agent with the same retriever and supervision, an always-on version of the same agent set, a hierarchical-audit baseline and a frozen frontier model, with 72 cases scored blind by trade-law assessors. Against the domain-adapted single agent, it raises citation support from 87.4% to 92.6%, a 5.2-point gain (95% bootstrap CI for the difference [3.6, 6.9]), lifts requirement-action coverage from 82.9% to 88.4%, reduces unsupported claims by 45.3%, and matches the always-on pipeline at 36.7% fewer tokens; blind expert scores are 4.16 of 5 against 3.60. Within the evaluated stack, interpretive capacity, not information supply, is what converts environmental regulation into sustainable business practice.

Keywords: sustainable business practices; emerging markets; sustainable development; environmental regulation; green product requirements; corporate environmental governance; digital innovation; artificial intelligence; WTO TBT Agreement; multi-agent systems

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1. Introduction

For most firms in emerging markets, global sustainability governance does not arrive as a climate agreement or a development target. It arrives as a product rule. Minimum energy performance thresholds, ecodesign and durability requirements, restricted-substance limits, recycled-content declarations and packaging marking obligations decide whether a consignment clears a border, and they have become the main channel through which environmental ambition is transmitted to the firms that manufacture traded goods. The volume is no longer marginal: the ePing platform received more than 7000 sanitary and phytosanitary and technical notifications in 2025, the highest annual figure since the transparency obligations took effect [1], and the platform records over 23,000 registered users [2]. That platform-wide volume indicates the scale of the transparency task; the empirical analysis below is confined to environment-related technical regulations and conformity assessment procedures. UNCTAD's 2026 assessment puts the economic weight plainly. Non-tariff-measure trade costs now exceed tariff costs in 88% of countries, requirements that are not properly notified impose a burden equivalent to a 28% tariff, and least developed countries forfeit around 10% of their exports to G20 markets because they cannot demonstrate compliance [3]. Technical barriers alone touch about 67% of trade by value [4].

"Green product requirements" is used here as an analytical umbrella rather than a category of WTO law. The corpus is confined to notifications classified as technical regulations, which lay down mandatory product characteristics, or as conformity assessment procedures, which determine how conformity is demonstrated [5]. Standards are voluntary unless a mandatory instrument incorporates them and are not sampled as stand-alone measures; sanitary and phytosanitary measures rest on a separate agreement and are excluded, as are non-tariff measures concerning neither environmental product characteristics nor their verification [4]. The categories differ in legal nature and in what an exporter must produce.

This creates a distributional problem at the heart of sustainable development. The binding constraint for an emerging-market exporter is rarely the absence of documents: notifications are public, searchable and, since the 2022 platform rebuild, filterable by sector and destination market [6]. The constraint is interpretation under resource scarcity. A single measure may exist as a base notification plus three addenda and a corrigendum, each with a different proposed entry-into-force date; the notified document may be a two-page summary pointing at a national gazette in a language the compliance officer does not read; the product scope may be omitted entirely. That ambiguity is expensive: stringent technical regulations act largely as a fixed cost and push exporters out of destination markets rather than merely shrinking their shipments [7], and standards depress the probability that firms in developing economies enter a market at all [8]. Large exporters absorb the fixed cost by staffing regulatory affairs functions. The small- and medium-sized enterprises that dominate emerging-market export bases do not [9], so a requirement written to raise environmental standards can end by excluding the firms least equipped to read it—an outcome that serves neither trade nor the environmental objective.

Emerging digital technologies are the obvious candidate for closing that capability gap, and the fit is not straightforward. Base notifications and their revisions share titles and product descriptions almost verbatim, so a language model tuned on similarity alone returns superseded text with high confidence, and models collapse the distinction between a proposed, an adopted and an actual entry-into-force date. A more consequential risk is unsupported assertion. Audited legal research tools hallucinate on a substantial share of queries even with retrieval present [10], and in a compliance workflow, an invented testing obligation wastes laboratory budget, while an invented exemption invites a border

rejection. Adding agents does not automatically repair this, since errors concentrate in identifiable agents and propagate downstream [11].

This paper develops and evaluates GRACE, a decision-support system that converts cross-border green product measures into evidence-grounded, human-reviewable compliance actions, so that a firm without a regulatory affairs department can act on environmental requirements with procedural assurance otherwise available mainly to firms carrying dedicated regulatory staff. Three commitments shape it. The deterministic parts of the task stay deterministic: symbols are normalised into notification families, temporal status is resolved into an explicit proposed/adopted/in-force triple, and product scope is reconciled before any agent is invoked. Retrieval is authority-aware, so a lower-ranked source can never carry a conclusion on its own. Orchestration is conditional, which keeps the system cheap enough for the organisations that serve these firms.

Evaluation runs on 214 held-out environmental TBT notification families drawn from the ePing SPS & TBT Platform and the WTO Environmental Database. Against the strongest single-agent baseline, which receives the same domain-adapted retriever and the same structured supervision, GRACE raises citation support from 87.4% to 92.6%, lifts requirement-action coverage from 82.9% to 88.4% and reduces unsupported assertions from 8.6% to 4.7%, while spending 36.7% fewer tokens than an always-on pipeline of the same agents.

Figure 1 sets out the problem and the boundary of the system: Figure 1a traces the path from the notification stream through the interpretive bottleneck to dated compliance work, and Figure 1b fixes what the system does and does not decide. GRACE is not offered as a new language model. Its domain-adaptation mechanisms—notification-family and date normalisation, and a graded authority hierarchy over source types—instantiate standard techniques for the WTO transparency record. What the paper claims as its methodological contribution is the evidence layer that follows them: a deterministic gate that withholds any obligation lacking admissible entailing support and re-emits it as an escalation question, together with the conditional orchestration that applies that check at a cost the target organisations can bear. The contribution is threefold. Empirically, the paper shows that the interpretive bottleneck between environmental regulation and firm action is measurable and can be substantially reduced within the evaluated corpus, and quantifies which mechanisms reduce it. Managerially, it demonstrates that regulatory sensing—the first step of any dynamic capability for sustainable internationalisation—can be industrialised at a cost a trade association or export promotion agency can bear. For governance, it identifies the notification metadata whose absence firms currently pay to reconstruct, which is an actionable design implication for the transparency instruments on which sustainable trade increasingly depends. Three questions organise the evaluation, one per mechanism: whether domain adaptation improves retrieval of the applicable record and its supporting evidence (RQ1); whether evidence grounding—separated verification and the gate—improves support and coverage beyond a single agent holding the same domain advantages (RQ2); and whether conditional routing preserves those gains at lower cost than an always-on pipeline (RQ3).

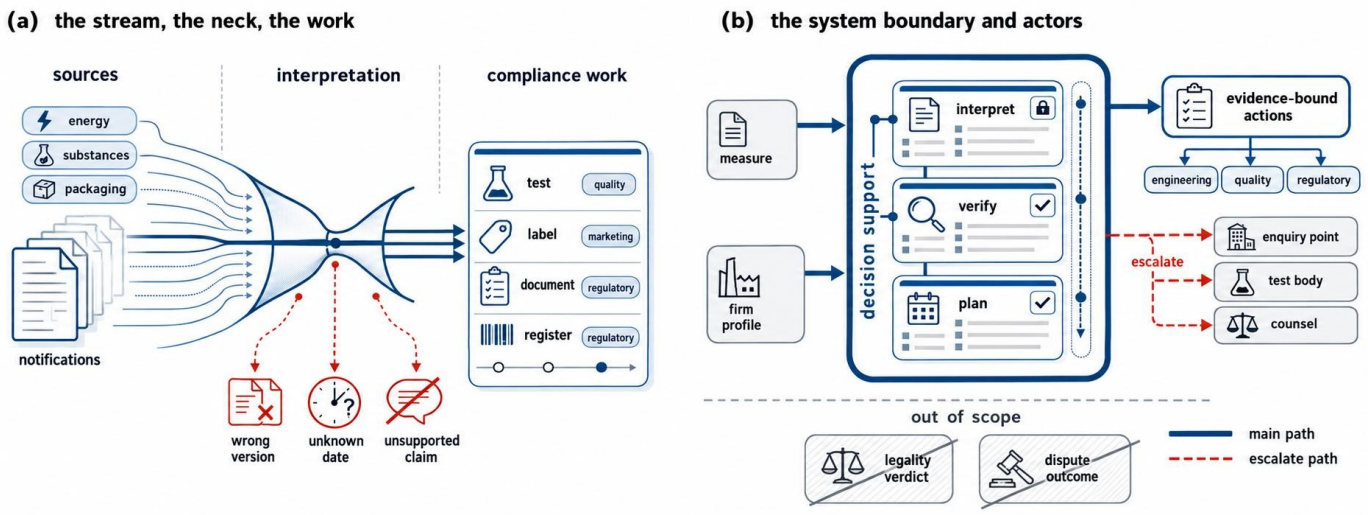


Figure 1. From the notification stream to firm action, and the boundary of the system. Panel (**a**): A stream of notified measures narrows through an interpretive bottleneck—where wrong-version, unknown-date and unsupported-claim errors arise—into dated compliance work owned by named functions. Panel (**b**): The system interprets, verifies and plans, hands evidence-bound actions to engineering, quality and regulatory owners, and escalates open questions; legality verdicts and dispute outcomes stay outside its scope. In capability terms, version, scope and date interpretation supports sensing, evidence-bound decision rules support seizing, and owner- and deadline-bound actions support transforming.

2. Related Work

2.1. Green Industrial Policy and the Product-Regulation Channel

Green industrial policy is usually debated through its fiscal instruments. Rodrik's case rests on the coordination and learning externalities that keep private investment in clean technology below the social optimum, and on the argument that the policy problem is one of institutional design rather than instrument choice [12]. Recent work sharpens the international dimension: states position domestic firms at particular points in global supply chains [13], and measures pursuing emissions, employment and supply-security objectives at once are difficult to reconcile with disciplines written for single-purpose instruments [14]. International business scholarship has begun to treat this as a first-order constraint on multinational strategy [15].

What this literature under-examines is the channel through which green industrial policy reaches firms most often. Energy performance thresholds, ecodesign requirements, substance restrictions and packaging obligations are not subsidies, yet they redirect investment, reshape supplier selection and dominate the WTO transparency record. The unit of analysis here is one such measure and the work it creates inside an exporting firm. This literature explains why states deploy green industrial policy; it does not explain how a published product rule becomes executable work for a firm that must comply with it.

2.2. Technical Barriers, Transparency and the Distribution of Compliance Burden

Quantitative trade research has established that technical measures operate through fixed rather than variable costs. Product standards reduce the number of firms that export and the number of destinations each serves, with effects concentrated at the extensive margin [16]; when one destination tightens, multi-destination firms reallocate towards markets without equivalent requirements [7]. For firms in developing economies, the entry effect dominates and falls hardest on those without dedicated regulatory capacity [8],

and survey evidence from manufacturing small- and medium-sized enterprises points the same way [17].

The WTO's institutional response has been transparency. Members must notify draft technical regulations and conformity assessment procedures that depart from relevant international standards and may significantly affect trade, under Articles 2.9.2 and 5.6.2 of the TBT Agreement [5], while Article 2.2 recognises environmental protection as a legitimate objective subject to a necessity test [18]. Concerns tabled at draft stage are markedly less likely to escalate into disputes [19], which locates the cheapest point of intervention early [9].

The gap this literature leaves is operational. A notification tells a compliance manager that a requirement exists, not which product lines it covers, which laboratory accreditation the destination market recognises, or how many weeks separate the comment deadline from the date her inventory becomes unsellable. UNCTAD's estimate that better regulatory transparency could cut non-tariff-measure trade costs by roughly 19% [3] is best read as an upper bound on what publication alone achieves. Transparency thus establishes availability without establishing interpretation.

2.3. Firm Capabilities for Regulatory Adaptation

Teece's formulation of dynamic capabilities as sensing, seizing and transforming provides the organising frame for that interpretive capacity [20]. Applied to environmental regulation, the empirical finding is that regulation translates into green practice only when firms commit resources and possess the entrepreneurial orientation to reconfigure products and processes [21]. Industry resistance to ecodesign standards stems less from the substance of requirements than from uncertainty about their interpretation and verification [22]; minimum energy performance standards deliver measurable efficiency gains alongside heterogeneous manufacturer responses [23], and extended producer responsibility fees across 25 European countries produced only small packaging reductions [24].

One feature of this evidence base shapes the present design: the binding uncertainty is procedural, since manufacturers generally know how to meet a threshold once they know which threshold applies, to which products, verified by whom and by when. A decision-support system should therefore be judged on whether it resolves those four questions with traceable evidence. Digital compliance tooling has moved faster than that evaluation, and the reported evidence concerns pipeline feasibility rather than grounding in authoritative sources [25,26]. The capability literature accordingly leaves the operational content of regulatory sensing unspecified, which in this setting reduces to four determinations: applicable version, product scope, operative date and admissible evidence.

2.4. Retrieval-Augmented Generation and Multi-Agent Orchestration

Retrieval-augmented generation grounds parametric knowledge in retrieved passages [27] and is now standard in legal applications, where lexical matching over statute numbers and citation strings remains indispensable alongside dense similarity [28]. Grounding is not the same as attribution. Frameworks that check whether each generated statement is entailed by its cited passage revealed substantial unsupported content even in retrieval-equipped systems [29]. Legal architectures have responded by structuring the source space: LegalGraphRAG organises authorities into a hierarchical graph and separates a researcher agent from an auditor that verifies retrieved support [30], while neuro-symbolic offloading compiles legal text into a typed representation executed deterministically [31].

Two features of the WTO setting break the assumptions these systems make. Their hierarchies encode precedential weight within a single legal order, whereas the transparency corpus mixes binding agreement text, notifications of measures that may never enter into

force, national implementing documents and committee statements. Their unit of retrieval is a stable document; here, it is a notification family whose members supersede one another over time.

Multi-agent research supplies the second half of the design. Verification-aware planning writes checkable conditions into the plan rather than validating after the fact [32], and process evaluation shows that scoring only final answers conceals skipped steps, fabricated tool calls and reliance on stale knowledge [33]. On the efficiency side, knowledge-graph-guided routing selects agents by task relationship instead of running a fixed topology [34], joint optimisation of prompts and topologies outperforms hand-designed pipelines [35], and a runtime filter that triggers a supervisor only when needed cuts token consumption on GAIA by 29.68% [36]. GRACE adopts conditional activation and verification-aware planning, instantiating the trigger conditions from regulatory signals. It deliberately does not adopt an adjudicator role, because the system produces neither legality findings nor dispute forecasts. The unresolved issue is that no existing architecture handles temporal notification families, graded source authority and conditional escalation within one evidence pipeline.

2.5. Conceptual Synthesis and Research Gap

The four literatures connect through a mediation argument. Green industrial policy reaches exporting firms largely through product rules, but publication does not itself create adaptive capability: uncertainty over version, scope, operative date and acceptable evidence interrupts regulatory sensing, and weak sensing delays seizing decisions and the transformation of specifications, supplier routines and technical documentation. Traceable interpretation is therefore treated here as a microfoundation of sustainable regulatory adaptation, with family, date and scope resolution supporting sensing, evidence gating disciplining seizing, and owner- and deadline-bound actions supporting transforming. The claim is bounded: decision support strengthens the informational and procedural antecedents of firm response, not sustainability outcomes directly, and each source of uncertainty is matched below to a mechanism, a comparator and a metric.

3. Materials and Methods

3.1. Problem Formulation and Design Requirements

The system receives a pair (m, p). The measure m is an environment-related technical regulation or conformity assessment procedure notified under the TBT Agreement, with the notified texts and official guidance it links to treated as evidence rather than as further sampling units; the firm profile p describes product type, target market, Harmonised System and International Classification for Standards codes, planned launch date and existing conformity documentation. The output is a compliance report $y = (\mathcal{M}, \mathcal{Q}, \mathcal{E}, \mathcal{A}, \mathcal{G}, \mathcal{U})$ comprising a measure card, a requirement map, an evidence pack, an action checklist, an information-gap list and an uncertainty and escalation record.

Four requirements constrain every design choice. Every substantive obligation retained in $\widehat{\mathcal{Q}}$ must bind to at least one passage in $\mathcal{E}$ at authority tier C or above. Every date in $\mathcal{A}$ must be labelled as proposed, adopted or in force, and must trace to the document that carries it. When evidence is insufficient, the entry moves to $\mathcal{G}$ and generates a question in $\mathcal{U}$. The report states no conclusion on the WTO consistency of the measure, no dispute outcome and no method of avoiding notification, testing, labelling or certification; requests of that kind are redirected to consultation with an enquiry point.

Figure 2 shows the resulting architecture: deterministic preprocessing, three specialised agents with fixed interfaces, a conditionally invoked evidence inspector, and an output validator.

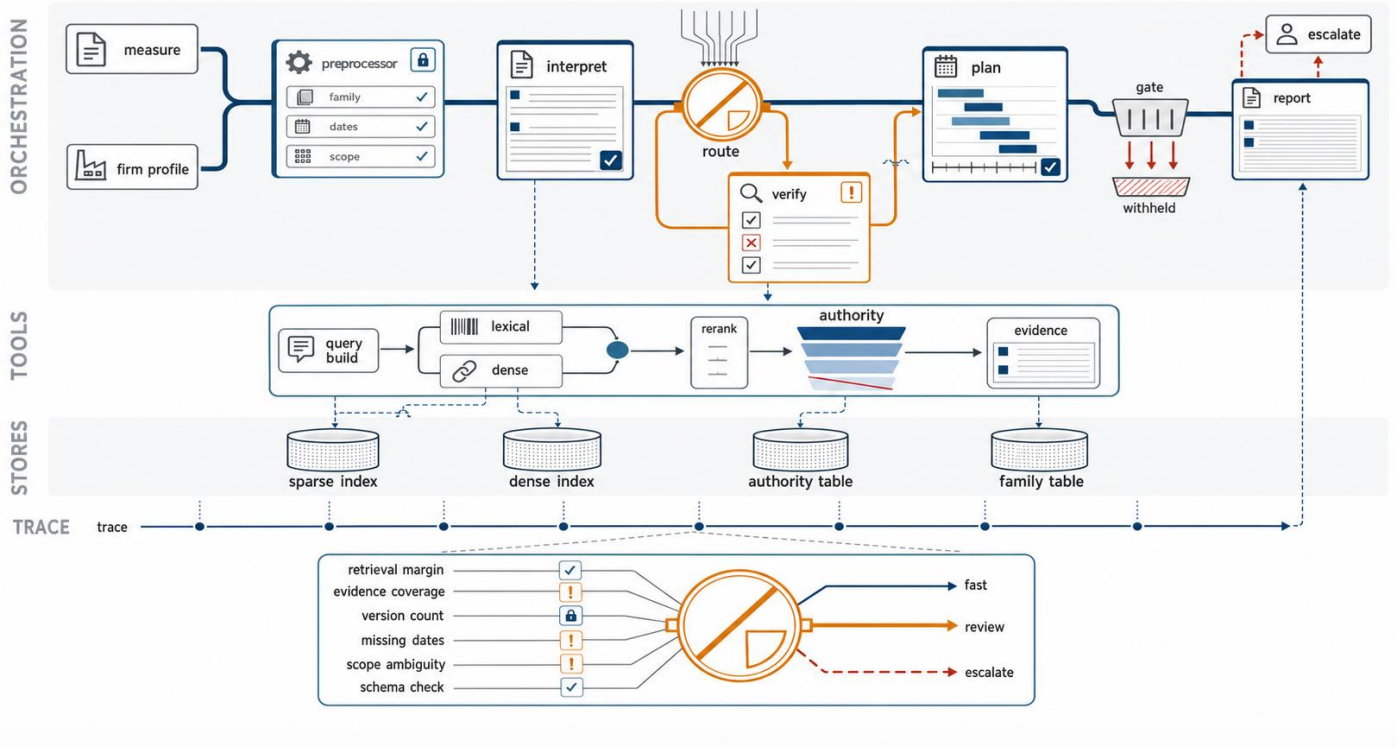


Figure 2. Architecture of GRACE. The orchestration layer runs deterministic preprocessing of families, dates and scope, then the measure interpreter, the conditionally activated verifier and the compliance planner, with the evidence gate and output validator guarding the report. The tools layer supplies hybrid lexical and dense retrieval, reranking and authority weighting over the stores. The lower panel shows the eight signals the router scores to assign a case to the fast, review or escalation path. Each layer addresses one failure mode: preprocessing addresses wrong-version and scope errors, retrieval and authority weighting address inadmissible support, inspection and the gate address unsupported assertion, and routing addresses the cost of applying those checks uniformly.

3.2. Deterministic Preprocessing

Preprocessing removes from the agents the work that admits an exact answer. WTO document symbols are parsed into a base symbol and a suffix drawn from the addendum, revision, corrigendum and supplement families, and the map $\mathcal{F}(d)$ assigns every symbol d to the notification family of its base. Within a family, the admissible document at reference time t is

$$d^{\star}(t) = \arg \max_{d \in \mathcal{F}^{-1}(f)} \{\delta(d) : \delta(d) \leq t\}, \tag{1}$$

where $\delta(d)$ is the circulation date of d. Documents circulated after t are retained but flagged, since a firm planning a launch needs to know that a further revision is pending. Where an addendum or corrigendum alters only part of the text it follows, the operative state of the family is the base text as amended, and superseded members are retained as context that cannot carry an obligation on its own. Dates are extracted into a triple $(t_{\text{prop}}, t_{\text{adopt}}, t_{\text{force}})$, with missing components represented explicitly and never imputed; the comment deadline is stored separately because it governs a different decision, namely whether the firm still has standing to comment.

Product scope is reconciled from three partially redundant signals. Harmonised System headings and International Classification for Standards codes are expanded to their prefix closures, and the free-text product description is matched against the firm profile with a lexical overlap score. When code-based and text-based scope disagree, or when both are absent, the case is marked scope-ambiguous, one of the router's trigger conditions. This

matters more than it might appear: in the study corpus, 31.3% of families carried no usable Harmonised System code.

3.3. Regulatory Measure Interpreter

The first agent emits a measure card under a fixed JSON schema. Eleven fields are populated: the document family and the admissible member selected by Equation (1), the notifying market, the product scope statements, Harmonised System and International Classification for Standards codes, the regulatory family, the instrument type, the requirement types, the comment deadline with the date triple, the version status, and a list of fields the agent could not determine. Every field is either populated from the source text with an evidence identifier or listed as uncertain; there is no third state and no field is imputed. Enumerated fields reject out-of-vocabulary values at decode time, and a validation failure triggers one repair attempt by the lightweight model before the case is routed to review.

Two distinctions inside the card do real work later. The first separates technical regulations from conformity assessment procedures, because redesigning a product to meet a threshold and arranging recognition of a test report are not interchangeable tasks. This operational distinction follows the legal boundary set out in Section 1. The second types each requirement as a testing, labelling, documentation, registration or design constraint. The typology is small on purpose: a finer taxonomy raised extraction disagreement during development without changing which actions the planner produced.

3.4. Authority-Aware Hybrid Retrieval

Retrieval combines a sparse and a dense channel because the two failure modes are complementary. Exact matching over document symbols, classification codes and legal terminology is a lexical problem; matching a requirement summary to the paragraph that supports it is a semantic one. Rankings from BM25 [28] and from the domain-adapted dense retriever are merged by reciprocal rank fusion [37],

$$s_{\mathrm{RRF}}(c) = \sum_{r \in \mathcal{R}} \frac{1}{\kappa + \mathrm{rank}_r(c)}, \tag{2}$$

with $\kappa = 60$ and $\mathcal{R}$ being the two channels. The fused top-k list is reranked, and the reranker score is then modulated by source authority and temporal admissibility

$$s(c) = \lambda_{a(c)} \cdot \mathbf{1}[c \preceq_t d^{\star}(t)] \cdot \tilde{s}_{\mathrm{rerank}}(c), \tag{3}$$

where $a(c) \in \{A, B, C, D\}$ is the authority tier of the passage, $\lambda_A = 1.00$, $\lambda_B = 0.92$, $\lambda_C = 0.78$, $\lambda_D = 0.45$, and the indicator suppresses passages belonging to superseded members of the family. Tier A covers agreement text and adopted committee decisions; tier B covers notifications, their revisions and the notified regulatory documents; tier C covers official explanatory material, implementation guidance and enquiry-point replies; tier D covers committee discussion and reliable secondary interpretation. The multiplicative form is deliberate. An additive bonus would let a large semantic score promote a tier-D passage above a tier-B one, which is exactly the substitution the system exists to prevent.

Figure 3 sets out the tier structure alongside the notification-family normalisation that Equation (1) implements.

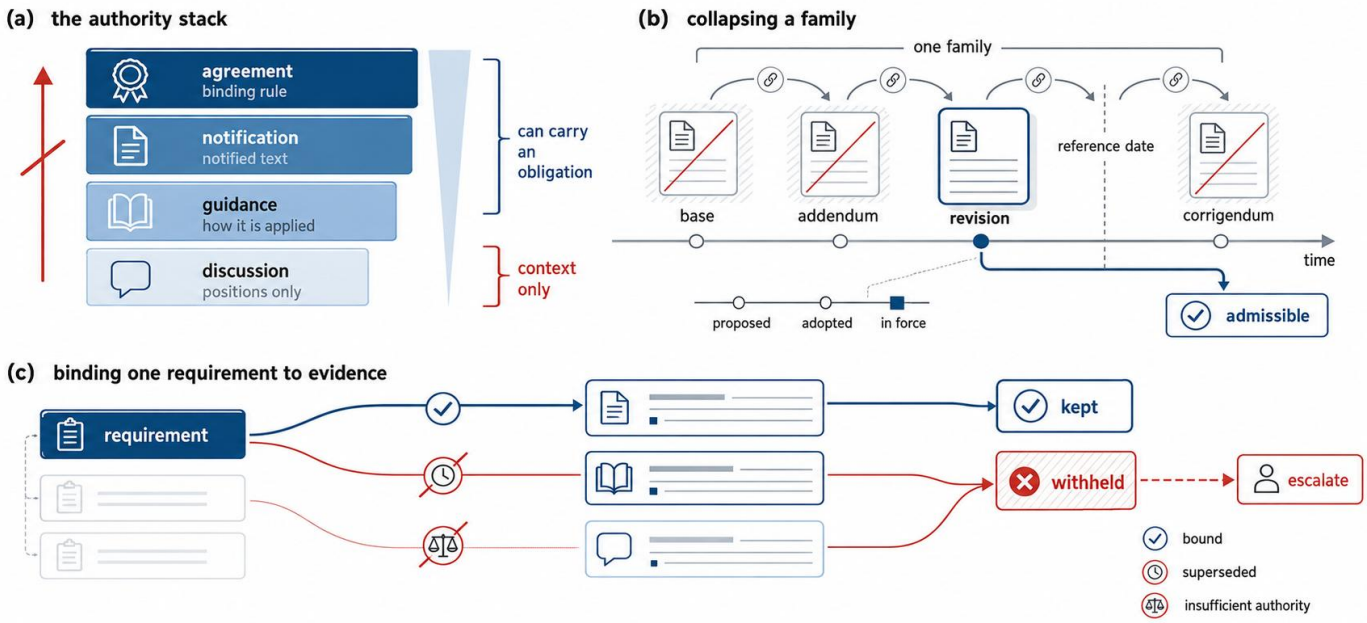


Figure 3. Source authority and notification-family normalisation. Panel (**a**): The four tiers, of which tiers A to C may support a retained requirement, while tier D is contextual and cannot carry an obligation on its own. Panel (**b**): A base notification, addendum, revision and corrigendum collapse into one family with a single admissible document at a reference date. Panel (**c**): A requirement with admissible support is kept; one resting on superseded or insufficient sources is withheld and escalated.

3.5. Evidence Inspector and the Evidence Gate

The inspector is not part of the default path. When invoked, it performs four checks: that each requirement in $\mathcal{Q}$ has support at tier C or above and none rests on tier D alone; that asserted dates match those carried by the retrieved documents; that no passage from a superseded family member is doing evidentiary work; and that no statement characterises a specific trade concern, a committee intervention or a draft measure as an established obligation, the most consequential category error available in this corpus.

The evidence gate then applies a deterministic rule to every requirement q with candidate evidence set E_q

$$\text{keep}(q) = \mathbf{1}\big[\, \exists\, c \in E_q : a(c) \preceq C \;\wedge\; \sigma(q, c) \geq \theta \,\big], \tag{4}$$

where $\sigma(q, c)$ is an entailment score between the requirement statement and the passage, and $\theta = 0.62$ was calibrated on the development split. Requirements failing the test are removed from $\mathcal{Q}$ and rewritten as items in $\mathcal{G}$ with an escalation question. Removal is not a fallback but the mechanism by which the coverage figures stay honest; Section 5.4 quantifies what happens when the gate is switched off.

3.6. Business Compliance Planner

The third agent maps surviving requirements onto firm actions: material and design verification, test and certification scheduling, labelling and packaging updates, technical file and supplier declaration preparation, the internal function accountable for each, and the calendar position of each action. Each action inherits the evidence identifiers of the requirement that generated it, so a compliance manager can trace any line of the checklist back to a paragraph in an official document. Actions whose feasibility depends on facts the system does not have are emitted as questions rather than as instructions.

Scheduling is where the temporal triple earns its cost. An action is anchored to t_{force} when that date is present, to t_{adopt} with an explicit provisional marker when it is not, and to the comment deadline when the measure is still at draft stage. Lead times are attached as ranges rather than points. The planner never compresses a schedule to fit a launch date; when the required actions do not fit before entry into force, it records the shortfall as an escalation item.

3.7. Adaptive Router

The router decides, per case, whether the inspector runs and whether the case escalates to a human. It scores a feature vector ϕ assembled entirely from signals available before the planner executes: top-1 reranker score, the margin between the first and second candidates, evidence coverage ratio, the number of distinct documents in the family, the count of missing components in the date triple, the scope-ambiguity indicator, the number of draft claims lacking evidence bindings and the schema validation result. A logistic model produces

$$u = \sigma\Big(w^{\top}\phi + b\Big), \qquad \text{route} = \begin{cases} \text{fast}, & u < \tau_1, \\ \text{review}, & \tau_1 \le u < \tau_2, \\ \text{escalate}, & u \ge \tau_2, \end{cases} \tag{5}$$

with $\tau_1 = 0.55$ and $\tau_2 = 0.86$ selected on the development split to hold expected citation support above 0.92 at minimum expected cost. A gradient-boosted tree and a 4B-parameter classifier were compared during development; the logistic model matched them within 0.4 percentage points and was retained because its weights are directly inspectable.

3.8. Domain Adaptation and the Reproducibility Boundary

All training material derives from existing public WTO records; no new dataset is constructed, named or released. The dense retriever is adapted contrastively, with positives formed from notification title and product description paired with the corresponding notified document, from requirement summaries paired with supporting official passages, and from base notifications paired with their family members. Hard negatives are the discriminations the task actually requires: same product with a different environmental objective, same objective with a different product, same market and year but an unrelated measure, and superseded versions of the correct document. Sampling is deterministic given a seed: each positive draws its seven negatives in the fixed composition, three same-objective, two same-product, one same-market-and-year and one superseded family member, restricted to the language of the positive and rejected above 0.8 token overlap. The objective is

$$\mathcal{L}_{\text{ret}} = -\log \frac{\exp(\text{sim}(q, c^{+})/\gamma)}{\exp(\text{sim}(q, c^{+})/\gamma) + \sum_{j=1}^{J} \exp\Big(\text{sim}(q, c_j^{-})/\gamma\Big)}, \tag{6}$$

with temperature $\gamma = 0.04$ and $J = 7$ hard negatives per positive.

Generation is adapted by four-bit quantised low-rank fine-tuning [38] on four task types only: measure-card extraction, requirement-to-evidence mapping, requirement-to-action mapping, and uncertainty and escalation generation. Supervision comes from the structured fields already present in the public records, with 312 high-value instances reviewed by hand, distributed across the four task types as 96 measure-card, 84 requirement-

to-evidence, 78 requirement-to-action and 54 uncertainty instances. The loss masks tokens outside the schema-relevant spans,

$$\mathcal{L}_{\text{gen}} = -\sum_{i=1}^{|y|} \mu_i \log P_\Theta(y_i \mid y_{<i}, x), \tag{7}$$

where $\mu_i = 1$ for tokens inside schema fields and evidence identifiers, and $\mu_i = 0.3$ elsewhere, which keeps the model fluent without letting narrative text dominate the gradient. Checkpoints are selected on the development split alone, and the evaluation families inform no threshold, checkpoint, prompt or router decision. Adapter weights and raw traces are not released; representative prompts, schemas, orchestration pseudocode, router coefficients, the split manifest and the pair-construction rules appear in Appendix B, which supports procedural replication and controlled reimplementation rather than bit-for-bit reproduction of the reported checkpoints.

4. Study Corpus and Experimental Setup

4.1. Public Sources and Internal Audit

Two public WTO sources supply the corpus, and both were audited locally on 12–13 August 2026. The ePing SPS & TBT Platform consolidates all sanitary and technical notifications submitted since 1995, together with specific trade concerns and enquiry-point contacts [39]; the downloaded extract contains 104,785 notification records across 34 columns, of which 64,932 are TBT records resolving to 60,452 distinct symbols after normalisation, with the most recent circulation date 12 August 2026. The WTO Environmental Database consolidates environment-related measures notified under the covered agreements, with harmonised objectives, measure types and sectors [40]. Its portal reported 10,291 environment-related notifications, 22,063 measures and 17,592 trade policy review entries when accessed on 13 August 2026 [41], and the measure count matches the workbook parsed locally.

Linking the two is the step that makes the corpus usable. Of the 7190 distinct TBT symbols in the Environmental Database, 7185 matched an ePing record exactly (99.93%); collapsing to families raised the match to 6914 of 6916, or 99.97%, the five unmatched symbols being corrigenda circulated outside the extract window. The Environmental Database therefore supplies harmonised environmental semantics and the ePing extract complete version, product and date structure, and the join is tight enough that neither must be trusted for what the other provides. Table 1 records the audit.

SPS records are removed before corpus construction, so the sampling pool contains TBT records only. Filtering by harmonised environmental objective, manufacturing or energy sector, and technical regulation or conformity assessment measure type yields 5128 records covering 5644 symbols and 5424 families. This is a research sampling pool, not a legal classification: inclusion carries no implication about the WTO consistency of any measure.

Table 1. Public sources, audited scale, use in the study, and the limits that constrain interpretation. All counts were obtained on 12–13 August 2026 and will drift as the platforms update.

Source	Audited Scale	Use in This Study	Interpretive Limits
ePing SPS & TBT Platform extract	104,785 records, 34 fields; 64,932 TBT rows; 60,452 distinct TBT symbols	TBT records only: version families, product scope, objectives, dates, notified-document links	A notification is not enacted law; proposed entry into force may differ from actual; product coverage may be under-specified

Table 1. *Cont.*

Source	Audited Scale	Use in This Study	Interpretive Limits
WTO Environmental Database workbook	22,063 measures, 21 fields; 6458 records labelled TBT; 7190 distinct TBT symbols	Harmonised environmental objective, measure type and sector semantics	Inclusion carries no implication as to consistency with WTO agreements; objectives and sectors are multi-label
Cross-source join	7185/7190 symbols (99.93%); 6914/6916 families (99.97%)	Combined semantic and temporal view of each family	Residual mismatches concentrate in corrigenda circulated near the extract boundary
TBT Agreement and committee material	Articles 2.2, 2.9, 5.6 and adopted transparency guidance	Tier-A authority for rule-level statements	Supplies the discipline, not the content of any national requirement
Notified regulatory documents and official guidance	Retrieved per case	Tier-B and tier-C evidence for specific obligations	Frequently non-English; translation is auxiliary, and the source link is always retained

4.2. Evaluation Cases

Training uses families circulated between 2009 and 2023 after the 66 held-out families are reserved; development uses families circulated during 2024. Evaluation uses 214 families held out entirely: 148 circulated in 2025 or 2026, and 66 reserved from the 2016–2023 period. Splitting is by family, never by record, so no revision of a training measure can appear at test time. Stratification spans three regulatory families—84 cases on energy efficiency and ecodesign, 76 on restricted substances, and 54 on packaging and circularity—and three evidence-difficulty tiers defined before any system was run: 96 single-version families with complete documents and unambiguous scope, 79 with multiple versions or partial documentation, and 39 missing final text, requiring cross-language reading, or with unresolved scope. The tier distribution is stratified rather than proportional. In the sampling pool of Section 4.1, 5644 symbols resolve to 5424 families, so multiple-version families are 4.1% of the population against 33.2% of the evaluation set; the design over-samples version multiplicity by about a factor of eight in order to obtain resolvable case counts in the tiers where the mechanisms under study operate. Reported means are accordingly conditional on this composition: the 214-family mean is a challenge-set estimate for mechanism testing, not a prevalence-weighted estimate for a randomly arriving notification. Routine performance is reported separately by tier in Section 5.6, and a deployment-weighted figure would require applying the full tier definition—missing final text and cross-language reading as well as version multiplicity—to the target stream. Each case is paired with a synthetic firm profile built from ordinary product categories.

4.3. Comparison Systems

Six comparators isolate distinct claims. B0 prompts the backbone with the measure text and no retrieval. B1 is generic retrieval-augmented generation with an off-the-shelf dense retriever and no authority or version handling. B2 is the load-bearing baseline: a single-agent system with the same domain-adapted retriever, the same hybrid fusion and the same structured supervision, differing only in that it has no separate inspector, no gate and no router. B3 runs the full agent set on every case. B4 implements hierarchical retrieval with post-retrieval audit in the manner of LegalGraphRAG [30], without family normalisation or adaptive routing. UB is a frozen frontier commercial model given the same hybrid retrieval, included as a capability reference. B2, B3 and GRACE share the corpus snapshot, retriever, generation adapter, schemas, decoding settings and evidence store, so what separates them is architecture rather than training or retrieval: B2 collapses interpretation, verification and planning into one pass without gate, escalation or router, and B3 invokes the same separated components as GRACE on every case. Table 2 records

which mechanism each system holds. The comparison against B2 carries the weight of the argument: it is the principal test of the incremental value of orchestration, since the single agent already carries every domain advantage.

Table 2. Mechanisms carried by each comparison system. A tick marks a mechanism in its GRACE form, an italic entry a different form, and a dash its absence. All systems share the evidence corpus snapshot, and B2, B3 and GRACE also hold the retriever, adapter, schemas and decoding constant, so the rows below are the only systematic differences among them.

Mechanism	B1	B2	B4	B3	GRACE
Domain-adapted retriever, hybrid fusion and reranking	—	✓	✓	✓	✓
Generation adapter and output schemas	—	✓	✓	✓	✓
Notification-family and date normalisation	—	✓	—	✓	✓
Source-authority weighting	—	✓	*graph*	✓	✓
Separate interpreter, verifier and planner	—	—	*audit*	✓	✓
Evidence inspector	—	—	*post hoc*	*always on*	*conditional*
Deterministic evidence gate	—	—	—	✓	✓
Escalation module	—	—	—	✓	✓
Adaptive router	—	—	—	—	✓

4.4. Metrics

Seven measures cover the pipeline, and two require the reference construction to be stated explicitly. Requirement-action coverage compares produced actions against a per-case checklist built from the notified document and its official implementation guidance, assembled before any system was run and fixed thereafter. The entailment score σ used in Equations (4) and (9) comes from a natural language inference (NLI) classifier: a DeBERTa-v3-large cross-encoder pre-trained on MNLI and fine-tuned on requirement-passage pairs from the development split, whose entailment probability is used directly as σ. Calibration uses 640 pairs drawn from development families only, so no evaluation family contributes a calibration pair. Two annotators with trade-compliance training labelled each pair independently as entailed or not entailed by the cited passage, with 58 disagreements resolved by a third; agreement before adjudication was $\kappa = 0.84$. The 640 pairs are divided into disjoint halves of 320: the gate verifier of Equation (4) is fine-tuned and thresholded on the first half, and the evaluation verifier is a second checkpoint trained under a different seed and calibrated on the second, so the two equations are scored by different checkpoints on disjoint calibration data. On its held-out half, the gate verifier reaches 0.91 accuracy against the adjudicated labels at $\theta = 0.62$, the value maximising F_1 over a 0.40 to 0.85 grid; no threshold was selected on the evaluation families.

Measure-card field-level macro-F1 scores extracted fields against the structured fields already present in the public records, which serve as reference without new annotation. Evidence recall@10 is the proportion of requirements whose supporting passage appears in the top 10 candidates, and current-version retrieval rate is the proportion of cases in which the admissible document under Equation (1) appears in the reranked evidence set. Citation support is measured over the statements a completed report makes, not over the requirements the gate retained. Writing $\widehat{\mathcal{Q}}$ for the requirements that survive Equation (4), the statement set is

$$\mathcal{S} \;=\; \widehat{\mathcal{Q}} \;\cup\; \mathcal{S}_{\text{act}} \;\cup\; \mathcal{S}_{\text{date}} \;\cup\; \mathcal{S}_{\text{scope}}, \tag{8}$$

which adds to the retained requirements the obligation-bearing clauses of the planner's action text, the dated assertions attached to each action, and the scope qualifications the planner introduces. Those three families are generated after the gate has run and are never presented to it. For $s \in \mathcal{S}$, let E_s be the evidence set carried by the report element in which s appears. Citation support rate is

$$\text{CSR} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \mathbf{1}\big[\, \exists\, c \in E_s : \sigma_{\text{eval}}(s, c) \geq \theta \;\wedge\; a(c) \preceq C \,\big], \tag{9}$$

so citation support is conditional on the statements a system emits and does not, on its own, measure completeness. The evaluation verifier σ_{eval} is a separate instantiation of the classifier that Equation (4) consults, initialised under a different seed, calibrated on the disjoint half described above and blind to the gate's decision, so the two equations quantify over different sets with different verifier instances. Two further measures complete the set. The unsupported-claim rate is the proportion of $s \in \mathcal{S}$ for which the NLI classifier finds no candidate in E_s that entails s at any authority tier. It is stricter than the complement of Equation (9), which also counts statements whose only support sits below tier C or below θ: a statement resting on a committee intervention is not citation-supported but is not an unsupported claim, whereas a fabricated testing obligation is both, and the gap between them is the share of output that is traceable but insufficiently authoritative. The completion rate is the proportion of cases returning a schema-valid report carrying all six components of y; incomplete cases are excluded from the per-statement measures and counted here. Cost is recorded per case as tokens, latency, agent invocations and money.

Because a system can raise citation support by declining weakly evidenced material, two further quantities accompany it wherever the gate is under examination. Requirement retention, $\text{RR} = \sum_i |\widehat{\mathcal{Q}}_i| / \sum_i |\mathcal{Q}_i^{\text{cand}}|$, is the share of drafted candidate requirements that survive Equation (4). Evidence-supported action coverage credits a reference action only when a matching action is produced and carries admissible entailing evidence,

$$\text{ESAC} \;=\; \frac{1}{|\mathcal{A}^\star|} \sum_{a^\star \in \mathcal{A}^\star} \mathbf{1}\Big[\, \exists\, \hat{a} \in \widehat{\mathcal{A}} : \text{match}(\hat{a}, a^\star) \;\wedge\; \exists\, c \in E_{\hat{a}} : \sigma_{\text{eval}}(\hat{a}, c) \geq \theta \;\wedge\; a(c) \preceq C \,\Big], \tag{10}$$

computed over matched actions rather than as a product of citation support and requirement-action coverage, whose denominators differ. Together, the three quantities separate selective reliability from completeness: what a system declines to state, how much of the reference work it reaches, and how much of that work it reaches with support that would survive inspection.

4.5. Implementation

Table 3 lists the configuration. The generation backbone is Qwen3.6-27B, a 27.78-billion-parameter dense model released under Apache 2.0 in April 2026 [42], used in text-only mode; routing, schema repair and date checking use Qwen3.5-4B [43]. Dense retrieval uses Nemotron-3-Embed-8B-BF16, which covers 34 languages [44], and reranking uses Qwen3-Reranker-4B [45]. Sparse and dense channels index the same 418,600 paragraph chunks under a FAISS HNSW structure. Retriever adaptation required 6.4 h and generation adaptation 19.3 h on four devices, and seeds 13, 29, 47, 61 and 83 govern data ordering, adapter initialisation and decoding.

Table 3. Model stack, adaptation settings and inference configuration. Sequence length is capped well below the backbone's native context because regulatory evidence is chunked at paragraph granularity, and longer windows increases latency without improving support.

Component	Setting	Value
Generation backbone	Model	Qwen3.6-27B (dense, 27.78B)
	Adaptation	4-bit QLoRA, rank 32, $\alpha = 64$
	Optimiser	AdamW, $\beta = (0.9, 0.95)$, weight decay 0.1
	Learning rate	1.2×10^{-4}, cosine, 3% warmup
	Schedule	3 epochs, effective batch 32, max length 16,384
Router and repair	Model	Qwen3.5-4B
	Router	Logistic, 8 features, $\tau_1 = 0.55$, $\tau_2 = 0.86$
	Repair budget	1 attempt per schema failure
Retrieval	Dense	Nemotron-3-Embed-8B-BF16, 4096-d
	Contrastive	$\gamma = 0.04$, 7 hard negatives, lr 2×10^{-5}, 2 epochs
	Sparse and fusion	BM25; reciprocal rank fusion, $\kappa = 60$
	Depth	$k = 20$ fused, reranked to 8
Entailment verifier	Model	DeBERTa-v3-large cross-encoder, MNLI-initialised
	Calibration	640 adjudicated pairs, development split, two disjoint halves of 320
	Instances	Gate and evaluation checkpoints, different seeds
Evidence handling	Authority weights	$\lambda_A = 1.00$, $\lambda_B = 0.92$, $\lambda_C = 0.78$, $\lambda_D = 0.45$
	Gate threshold	$\theta = 0.62$
	Chunking	Paragraph, 320 tokens, 48-token overlap
Execution	Hardware	4 × NVIDIA A100 80 GB
	Decoding	Temperature 0.2, top-p 0.9, schema-constrained
	Seeds	13, 29, 47, 61, 83

4.6. Expert Assessment Protocol

Three assessors with backgrounds in international economic law and trade compliance took part: two scored every case independently and a third arbitrated disagreements without seeing the first two scores. The primary assessors are a trade-compliance adviser with eleven years of practice and a legal academic specialising in technical-barrier disciplines with nine; the arbitrator, a former enquiry-point officer, scored only referred judgements. None took part in building any system or are affiliated with the funding body. All worked from four common calibration examples, saw the evidence passages and source metadata, and scored the delivered report rather than predicted adoption. The two primary assessors scored 72 cases sampled to preserve the regulatory-family and difficulty-tier proportions of the full set. Reports from B2, B3 and GRACE were presented in randomised order with system identity concealed, and six dimensions were scored on a five-point scale. Disagreements exceeding one point were referred to the arbitrating assessor, which occurred on 41 of 1296 judgements; quadratic-weighted Cohen's κ was 0.71 overall. No language model participated in scoring.

Statistical comparisons use paired bootstrap confidence intervals with 10,000 resamples for the automatic metrics, Wilcoxon signed-rank tests for expert scores, McNemar's test for binary case-level outcomes and Holm correction across the seven metric families. Effect sizes are reported as Cliff's δ. The notification family is the primary resampling unit: families are drawn with replacement, their five seed outputs resampled at the second level, and every system is scored on identical indices so outputs stay paired. Table 4 and the paired comparisons reported in Section 5.2 use the same logic, and significance is read from the paired differences rather than from overlap between absolute intervals.

Table 4. Main comparison on 214 held-out notification families. Entries are means with 95% paired hierarchical-bootstrap confidence intervals in brackets, from 10,000 resamples with the notification family as the primary unit; seed-level dispersion is reported in Appendix B.3. Higher is better except for the unsupported-claim rate. Best value in each column is in bold. UB is a frozen frontier commercial model provided with the same hybrid retrieval and is reported as a capability reference, not as a component of the proposed system.

Panel A. Extraction and retrieval.

System	Card F1	R@10	Current Ver.
B0 Direct LLM	61.4 [59.5, 63.2]	—	22.8 [20.5, 25.2]
B1 Generic RAG	85.7 [84.7, 86.6]	76.4 [74.8, 78.1]	80.9 [79.1, 82.7]
B2 Domain single-agent	91.6 [90.9, 92.3]	84.8 [83.7, 85.9]	92.1 [91.1, 93.0]
B4 Hierarchical audit	91.9 [91.1, 92.7]	85.6 [84.4, 86.7]	93.4 [92.4, 94.4]
B3 Fixed multi-agent	92.4 [91.8, 93.0]	86.1 [85.1, 87.1]	94.7 [93.9, 95.5]
UB Frontier + hybrid	90.8 [90.1, 91.5]	85.2 [84.2, 86.2]	89.6 [88.5, 90.7]
GRACE	**93.1 [92.5, 93.7]**	**87.3 [86.4, 88.2]**	**95.8 [95.1, 96.5]**

Panel B. Reliability, action coverage and completion.

System	Citation Supp.	Unsupp. Claims	Action Cov.	Completion
B0 Direct LLM	31.6 [29.3, 34.0]	41.7 [39.0, 44.4]	48.3 [46.0, 50.7]	88.1 [86.0, 90.0]
B1 Generic RAG	79.2 [77.7, 80.7]	15.3 [14.0, 16.7]	74.6 [72.9, 76.2]	92.8 [91.5, 94.0]
B2 Domain single-agent	87.4 [86.2, 88.6]	8.6 [7.8, 9.5]	82.9 [81.6, 84.2]	96.2 [95.3, 97.0]
B4 Hierarchical audit	89.8 [88.7, 90.9]	6.9 [6.1, 7.7]	84.5 [83.1, 85.8]	96.6 [95.7, 97.4]
B3 Fixed multi-agent	91.3 [90.4, 92.2]	5.4 [4.8, 6.1]	87.1 [86.0, 88.2]	97.1 [96.4, 97.8]
UB Frontier + hybrid	88.1 [87.1, 89.1]	7.8 [7.0, 8.7]	85.7 [84.5, 86.9]	97.4 [96.7, 98.1]
GRACE	**92.6 [91.7, 93.5]**	**4.7 [4.2, 5.3]**	**88.4 [87.4, 89.4]**	**97.9 [97.3, 98.5]**

5. Results

5.1. Domain Optimisation Is Necessary and Insufficient

Does adapting retrieval and interpretation to WTO structure improve regulatory output? Table 4 and Figure 4 report the corresponding comparison. Moving from generic retrieval-augmented generation to the domain-adapted single agent raises measure-card macro-F1 from 85.7% to 91.6%, evidence recall@10 from 76.4% to 84.8% and citation support from 79.2% to 87.4%, while unsupported claims fall from 15.3% to 8.6%. The largest single gain is current-version retrieval, from 80.9% to 92.1%: a generic retriever ranks a superseded revision and its replacement almost identically because they differ by a few sentences, whereas Equation (1) removes the superseded member from contention before scoring begins.

Direct prompting without retrieval is not competitive on any regulatory dimension, and its 41.7% unsupported-claim rate confirms that parametric knowledge of a daily-updated corpus is no substitute. The frozen frontier model is more informative: given the same hybrid retrieval, it reaches 88.1% citation support and 85.7% action coverage, ahead of the domain-adapted single agent on coverage but 2.5 points behind on current-version retrieval. Raw capability compensates for missing domain structure on tasks that reward reading comprehension, but not on the task that requires knowing that a symbol ending in Rev.2 supersedes one ending in Add.1.

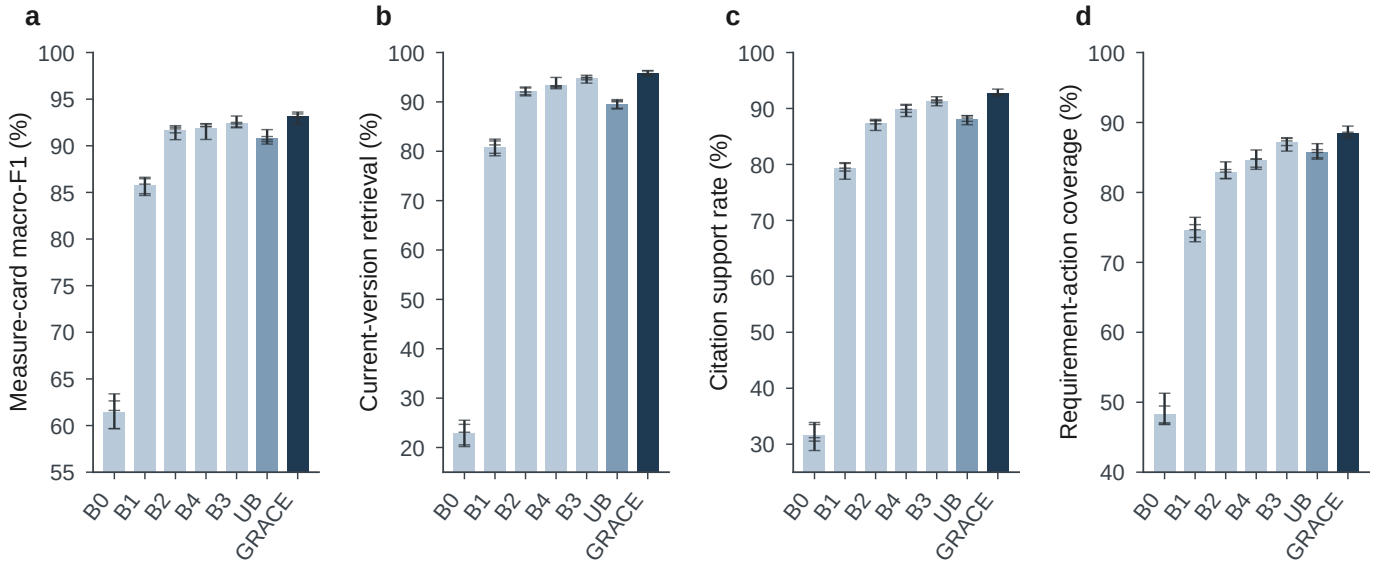


Figure 4. Main comparison across four evaluation axes. Whiskers span the observed range across the five seeds, and short rules mark individual seed values, so they are asymmetric about the mean; seed dispersion is reported in Appendix B.3. (**a**) Measure-card macro-F1. (**b**) Current-version retrieval rate. (**c**) Citation support after the evidence gate. (**d**) Requirement-action coverage.

5.2. What the Multi-Agent Configuration Adds

Given a strong domain-adapted single agent, what does separating interpretation, retrieval, verification and planning buy? Because B2 holds the corpus, retriever, adapter, schemas and decoding constant (Table 2), the contrast isolates orchestration rather than domain adaptation. The answer is concentrated in the reliability metrics. GRACE improves on B2 by 5.2 points of citation support, with a bootstrap interval of $[3.6, 6.9]$ and $p < 0.001$ after Holm correction, and by 5.5 points of requirement-action coverage. Unsupported claims fall by 3.9 points, a 45.3% relative reduction, while measure-card F1 moves by only 1.5 points. The pattern is coherent: field extraction from structured notifications is close to saturated once the retriever is adapted, so the additional agents contribute where a single pass has no natural checkpoint, namely in deciding whether an asserted obligation survives contact with its evidence.

Figure 5 reports the full test battery. Every comparison against B2 clears the Holm-corrected threshold with effect sizes between 0.36 and 0.58, and comparisons against the always-on pipeline B3 are smaller but consistently favourable, at 1.3 points on both citation support and action coverage.

5.3. Routing Converts Evidence Difficulty into Compute

Does conditional activation deliver the efficiency it promises? Table 5 and Figure 6 show that it does. GRACE invokes 2.64 agent calls per case against the fixed pipeline's four, consuming 21.9 thousand tokens against 34.6 thousand and completing in 20.8 s against 31.8: reductions of 34.0% in agent calls, 36.7% in tokens and 34.6% in latency, obtained while citation support rises by 1.3 points.

The tier-level breakdown explains where the saving originates. On tier 1, where the family has a single version and scope is unambiguous, 87.5% of cases take the fast path, and consumption falls to 15.8 thousand tokens, 52% below the fixed pipeline on the same cases; on tier 3, the fast-path share drops to 10.2%, the inspector fires on 74.4% of cases and 15.4% escalate, so consumption rises to 31.7 thousand, only 17% below. Effort tracks difficulty, and Table 6 shows the consequence for the distribution of spend: tier 1 is 44.9% of the corpus and absorbs 32.3% of the adaptive token budget against 42.6% of the always-on pipeline's,

while tier 3 is 18.2% and absorbs 26.4% against 20.0%. Conditional activation moves budget to the cases that need it; the always-on pipeline spends in proportion to volume.

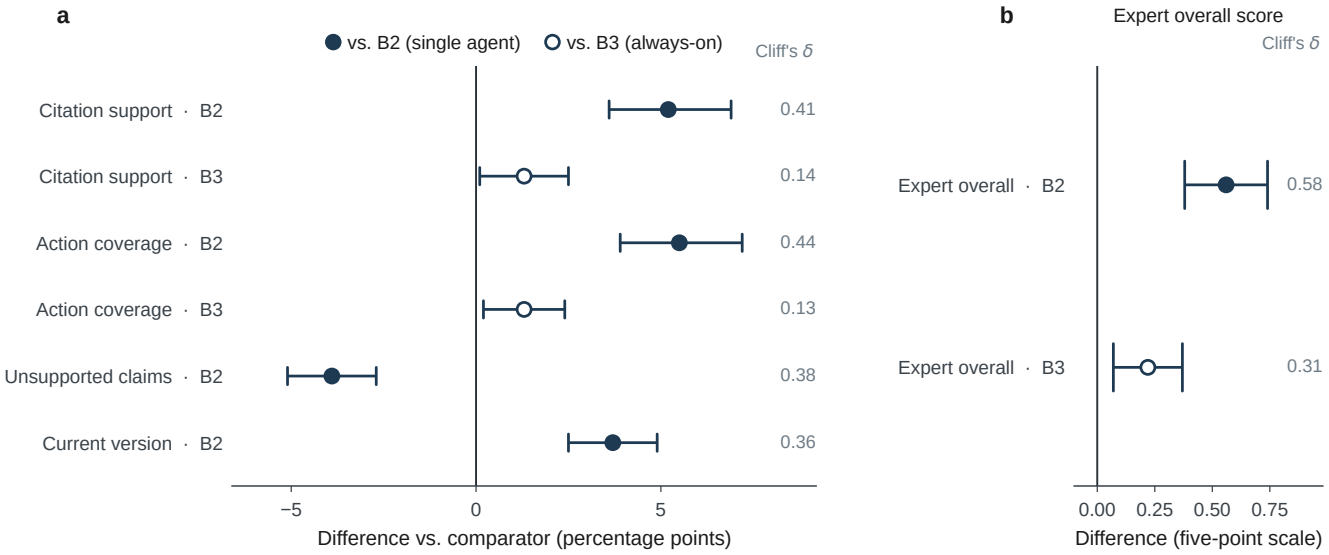


Figure 5. Paired statistical comparisons on the 214-family evaluation set. Points are the observed difference and bars the bootstrap percentile interval over 10,000 resamples; filled points contrast GRACE with B2 and open points with B3, and Cliff's δ is printed at the right of each row. (**a**) Automatic metrics, in percentage points; the unsupported-claim contrast is negative because a reduction is an improvement. (**b**) Expert overall score, plotted separately because the unit differs. All intervals exclude zero; Holm-corrected p-values are <0.001 for every contrast against B2, 0.034 and 0.041 for the citation-support and action-coverage contrasts against B3, and 0.008 for the expert contrast.

The comparison against the frontier model is the more relevant cost contrast for deployment. It runs a single pass and consumes fewer tokens, but at published API pricing, it costs USD 0.412 per completed case against USD 0.047 for GRACE on the local stack—a factor of 8.8—while scoring lower on all six regulatory metrics. For a firm screening several hundred notifications a quarter, that is the difference between a routine process and a budget line requiring approval.

Table 5. Computational cost and routing behaviour per completed case, averaged over the 214 evaluation families and five seeds. Monetary cost for GRACE and the open-weight baselines, reflecting amortised GPU time on the deployment stack; the frontier reference reflects published API pricing at the time of the experiments.

System	Tokens (k)	Latency (s)	Agent Calls	Retrieval Calls	USD/Case
B0 Direct LLM	3.1	4.2	1.00	0.0	0.008
B1 Generic RAG	11.8	9.6	1.00	1.0	0.026
B2 Domain single-agent	14.3	12.4	1.00	1.0	0.031
B4 Hierarchical audit	26.7	24.1	3.00	2.0	0.058
B3 Fixed multi-agent	34.6	31.8	4.00	2.0	0.074
UB Frontier + hybrid	16.2	14.9	1.00	1.0	0.412
GRACE	21.9	20.8	2.64	1.43	0.047

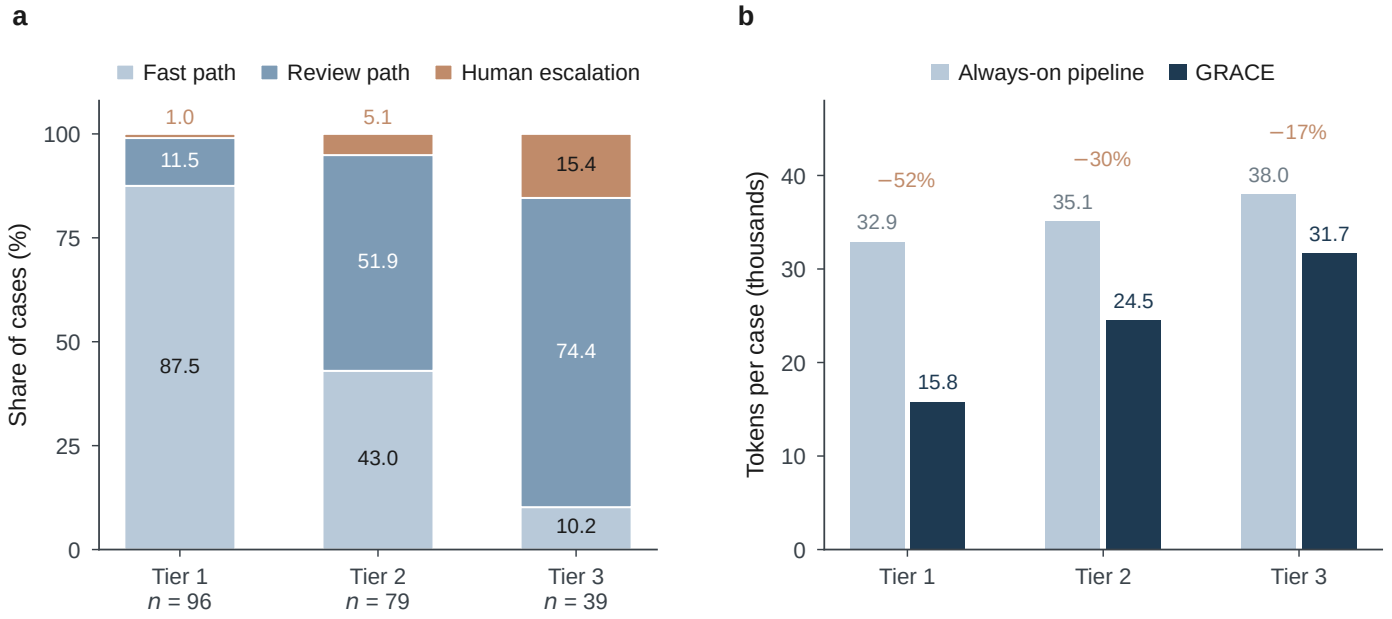


Figure 6. Routing behaviour across the three evidence-difficulty tiers defined before any system was run. (**a**) Share of cases taking the fast path, the review path and human escalation. (**b**) Tokens per case against the always-on pipeline on the same cases, with the reduction printed above each pair. Numerals inside the bars are percentages of cases; tier sizes are given on the axis.

Table 6. Where the token budget is spent. Tier totals are the tier mean times the tier size, and the corpus means reproduce the entries of Table 5. Share is the percentage of the system's total consumption.

Stratum	*n*	GRACE		Always-on Pipeline	
		Tokens (k)	Share (%)	Tokens (k)	Share (%)
Tier 1	96	1516.8	32.3	3158.4	42.6
Tier 2	79	1935.5	41.3	2772.9	37.4
Tier 3	39	1236.3	26.4	1482.0	20.0
Corpus total	214	4688.6	100.0	7413.3	100.0
Per case	—	21.9	—	34.6	—

5.4. Ablation

Which components carry the result? Table 7 and Figure 7 remove one at a time. Retriever domain adaptation is the largest contributor to evidence quality: without it, recall@10 falls by 8.4 points to 78.9%, because generic embeddings cannot separate two notifications that share a product category and differ only in environmental objective. Notification-family normalisation is the largest contributor to temporal correctness, and its removal costs 14.2 points of current-version retrieval, the single largest movement anywhere in the study. Removing the authority hierarchy leaves retrieval almost intact but raises unsupported claims from 4.7% to 8.4%, since committee discussion and secondary interpretation begin to carry conclusions that only agreement text or notified documents should carry.

The evidence gate produces the most instructive result. Switching it off raises requirement-action coverage from 88.4% to 89.6%, while citation support falls by 8.0 points and unsupported claims more than double to 11.8%, because ungrounded requirements are then counted alongside grounded ones. A system evaluated on coverage alone would therefore appear to improve by removing the component that makes its output trustworthy, which is a strong argument for reporting support and coverage together. Table 8 and Figure 8 therefore sweep the threshold with every other component fixed, reading support jointly with retention. At the selected threshold, the gate retained 2183 of 2441 drafted

requirements, a retention rate of 89.4%; tightening it to 0.80 buys 2.6 further points of support for 9.6 points of retention and 5.4 of coverage, while the development-selected $\theta = 0.62$ also yields the highest evidence-supported action coverage, at 81.9%, among the swept settings; no threshold was retuned on the held-out evaluation set. Completion moves by 0.2 points across the sweep, since withheld requirements are re-emitted as information gaps rather than dropped.

Table 7. Ablation of the six mechanisms, evaluated on the same 214 families. All values are percentages except tokens per case. The In B2 column marks the three mechanisms the single-agent baseline also carries, so ablating those measures their absolute contribution; the remaining three rows are selected GRACE-specific mechanisms rather than the full set listed in Table 2. The evidence-gate row is the one case where a metric improves under ablation.

Configuration	In B2	Card F1	R@10	Current Ver.	Citation Supp.	Unsupp. Claims	Action Cov.	Tokens (k)
Full GRACE	—	93.1	87.3	95.8	92.6	4.7	88.4	21.9
w/o retriever adaptation	yes	92.3	78.9	90.2	86.9	7.9	83.1	22.4
w/o authority hierarchy	yes	92.8	85.4	93.1	87.2	8.4	87.6	21.5
w/o family normalisation	yes	90.4	84.7	81.6	88.3	7.1	85.8	20.8
w/o evidence gate	no	93.0	87.1	95.3	84.6	11.8	89.6	19.6
w/o adaptive router	no	92.4	86.1	94.7	91.3	5.4	87.1	34.6
w/o escalation module	no	92.9	87.0	95.4	91.8	6.2	88.0	21.4

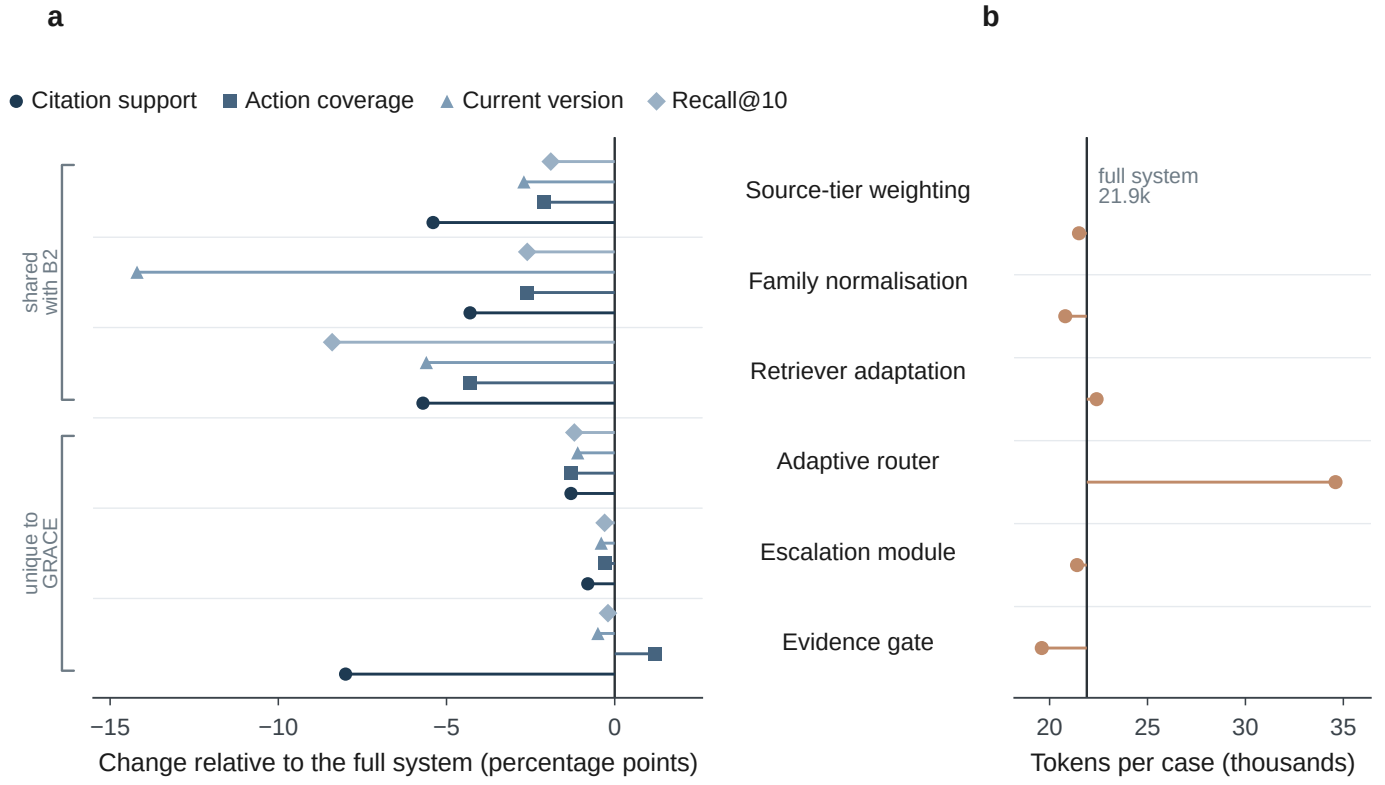


Figure 7. Ablation of the six mechanisms, one removed at a time, sharing a common row for each mechanism. The bracket on the left separates the three mechanisms that the single-agent baseline B2 also carries from three selected GRACE-specific mechanisms. (**a**) Change relative to the full system across four quality metrics, in percentage points; marker shape identifies the metric, and the vertical rule is the full system. (**b**) Token consumption per case on the same rows, measured against the full system at 21.9 thousand; only the router carries a substantial cost function.

Table 8. Sensitivity of the joint reliability and completeness profile to the evidence-gate threshold, with the retriever, router, model stack and seeds held fixed. RR is requirement retention, CSR is citation support, RAC is requirement-action coverage and ESAC is evidence-supported action coverage. The selected operating point is in bold; higher is better except for the unsupported-claim rate.

Gate Setting	RR	CSR	RAC	ESAC	Unsupp. Claims	Completion
No gate	100.0	84.6	89.6	75.8	11.8	97.9
$\theta = 0.45$	96.6	87.2	89.4	78.0	9.2	97.9
$\theta = 0.55$	92.7	90.8	89.0	80.8	6.2	97.9
$\theta = 0.62$	**89.4**	**92.6**	**88.4**	**81.9**	**4.7**	**97.9**
$\theta = 0.70$	85.7	94.0	86.8	81.6	3.8	97.8
$\theta = 0.80$	79.8	95.2	83.0	79.0	2.9	97.7

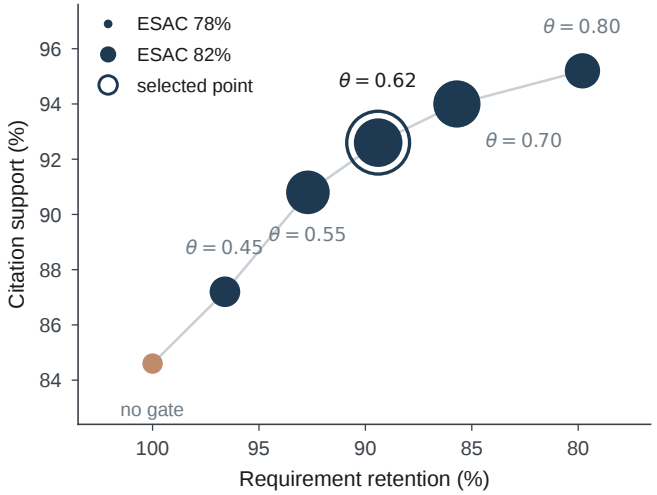


Figure 8. The evidence gate as a frontier rather than a switch: citation support against requirement retention across the sweep of Table 8, with marker area giving evidence-supported action coverage and the ring marking the selected point $\theta = 0.62$.

5.5. What Actually Triggers Review

Section 5.3 shows that conditional activation saves compute. It does not show what the router is responding to. Figure 9 answers that by displaying every case against the subset of the eight routing conditions it activates, using a set-intersection matrix in the manner of Lex et al. [46]. Fourteen distinct activation patterns account for 199 of the 214 cases; the remaining 15 are pooled into a single column.

The most common condition of a case is no condition at all. Sixty-one cases, 28.5% of the corpus, activate nothing and route fast at a median score of 0.15. That figure is the practical ceiling on what any conditional-activation scheme can save.

Version multiplicity is a weak signal, which inverts the natural expectation. Multi-version families are the most frequent condition at 71 cases, yet the 27 that activate it alone route entirely to the fast path: Equation (1) resolves supersession before the router runs, so a family with four members carries no more ambiguity than one with a single member. A system treating document count as a proxy for difficulty would send all 71 to review.

Escalation is combinatorial, and one condition sits underneath all of it. Ten of the eleven escalated cases arise from patterns activating four or more conditions at once, and every escalating pattern includes incomplete evidence coverage; narrow retrieval margins appear alongside it but never escalate alone. The thresholds partition the observed patterns almost cleanly, so the routing decision is close to a property of the pattern itself.

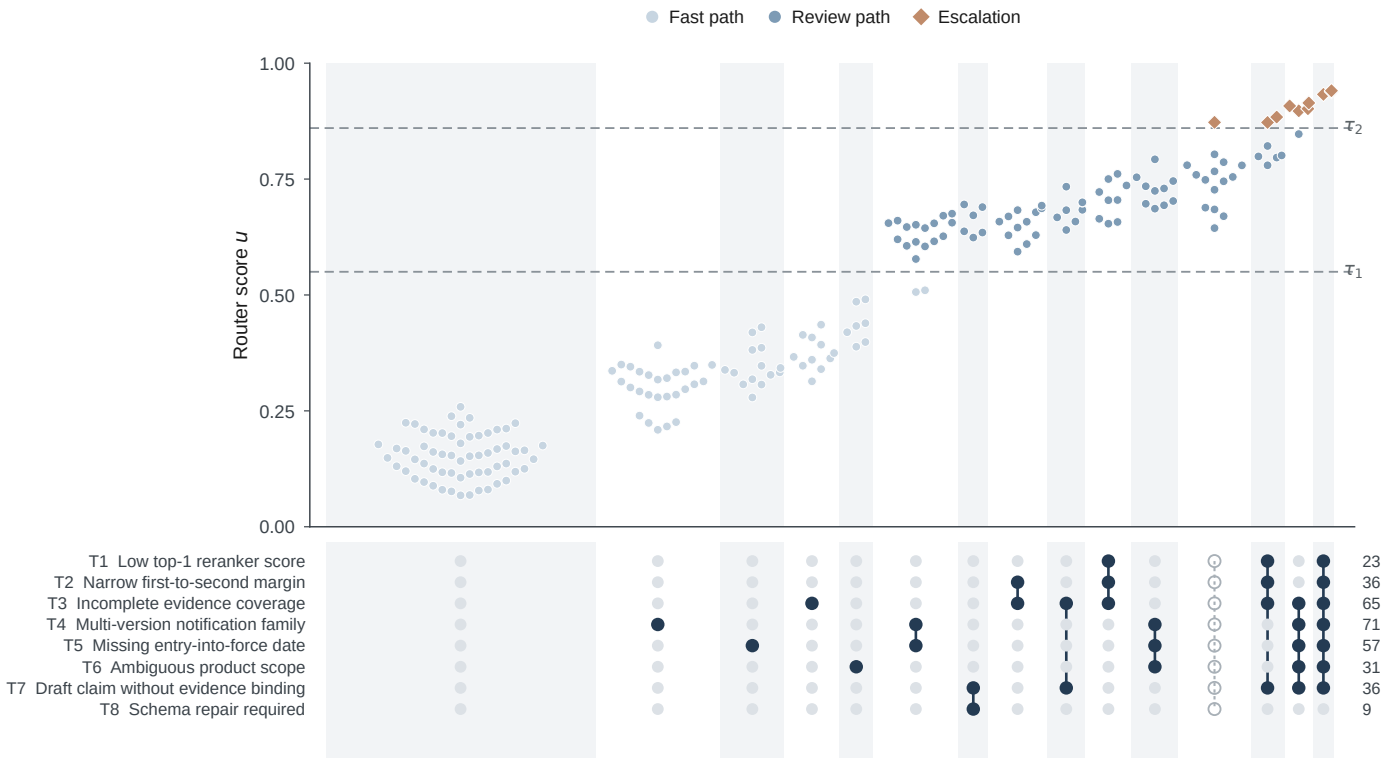


Figure 9. Router conditions activated by each of the 214 evaluation cases. The lower panel is a set-intersection matrix: rows are the eight routing conditions, columns the observed activation patterns, a filled dot marks an active condition, and the numerals at the right give each condition's marginal count. Column width is proportional to the number of cases sharing the pattern. The upper panel places every case at its router score u, coloured by the path it received, with dashed rules at $\tau_1 = 0.55$ and $\tau_2 = 0.86$; the rightmost column pools 15 cases whose patterns each occur fewer than four times.

5.6. Stability Across Regulatory Families, Time and Hyperparameters

Do the gains survive disaggregation? Table 9 reports results by regulatory family and by temporal slice. Citation support for GRACE ranges from 91.0% on packaging and circularity to 93.8% on energy efficiency and ecodesign, and the margin over the domain-adapted single agent stays between 4.9 and 5.4 points. Packaging and circularity is the hardest family for every system, which follows from the structure of the requirements: recycled-content and recyclability obligations more often reference national implementing instruments not included in the notification, so evidence must be assembled from tier-C material rather than read from the notified document.

The temporal split shows the expected direction and a modest magnitude. On families circulated in 2025 and 2026, GRACE reaches 92.2% citation support against 93.4% on held-out earlier families, while the single-agent system drops 1.7 points over the same contrast. Newer measures are harder for both because notified documents are more often still provisional.

Table 10 disaggregates quality by evidence-difficulty tier, since the set over-samples difficult families. On tier 1—the routine condition—margins of 4.7 points of citation support and 5.1 of action coverage persist where the mechanisms have least to resolve, so the advantage is not an artefact of the hard strata. Absolute performance then falls across tiers, while the margin widens to 6.2 points on tier 3, the stratum with the highest escalation rate and therefore the boundary on autonomous use.

Table 9. Disaggregated results by regulatory family and by temporal slice. Family sizes sum to the 214 evaluation cases, and the weighted means reproduce the corresponding entries in Table 4. CSR denotes citation support rate, and RAC denotes requirement-action coverage.

Stratum	*n*	B2 Domain Single-Agent		GRACE		Δ CSR
		CSR	RAC	CSR	RAC	
Energy efficiency and ecodesign	84	88.9	84.7	93.8	90.1	+4.9
Restricted substances and materials	76	87.1	82.6	92.4	88.3	+5.3
Packaging, waste and circularity	54	85.6	80.8	91.0	86.2	+5.4
Held-out 2016–2023 families	66	88.6	84.0	93.4	89.5	+4.8
Out-of-period 2025–2026 families	148	86.9	82.4	92.2	87.9	+5.3

Table 10. Quality by evidence-difficulty tier, on the same 214 families and five seeds. Tier weights reproduce the corpus means of Table 4 and the token means of Table 5. Tier 1 is the routine condition: a single-version family with complete documents and unambiguous scope.

Tier	System	CSR	RAC	Unsupp.	Completion	Tokens (k)
Tier 1 ($n = 96$)	B2	89.5	85.6	7.2	97.2	12.7
	B3	93.1	89.3	3.9	98.1	32.9
	GRACE	94.2	90.7	3.0	98.7	15.8
Tier 2 ($n = 79$)	B2	87.2	82.9	8.6	96.3	14.9
	B3	91.3	87.3	5.2	97.2	35.1
	GRACE	92.5	88.5	4.6	98.1	24.5
Tier 3 ($n = 39$)	B2	82.7	76.3	12.0	93.6	17.2
	B3	86.9	81.3	9.6	94.5	38.0
	GRACE	88.9	82.4	9.0	95.6	31.7

What marginal gain does each control still provide at the selected operating point? Figure 10 answers this by drawing, at each of the 20 configurations, what the next step of each control gains and what it costs: arrow length is the citation support added, and arrow thickness is the extra tokens. Raising the router review threshold τ_1 from 0.35 to 0.45 at $k = 20$ adds 2.2 points of support for 3.2 thousand tokens; the same-sized step from 0.65 to 0.75 adds 0.1 points for 6.4 thousand, a 40-fold deterioration in return per token. Retrieval depth behaves the same way at $\tau_1 = 0.55$. The selected configuration is the first grid point at which both controls fall below half a point of support per step, and the field is flat beyond it.

5.7. Selective Reliability and the Value of Declining

Across the 214 families, the interpreter extracted 2441 candidate requirements. The gate admitted 2183 to the action checklist, converted 171 into information gaps and 87 into escalation questions, so 10.6% of what the system drafts is withdrawn before it reaches the user. Selective withholding is preferable to assertion when evidentiary sufficiency is not met, provided the declined cases are the ones the system would otherwise get wrong. They are: the router escalated 11 cases, or 5.1%, and on the 203 retained cases citation support rises to 93.8%, while unsupported claims fall to 3.9%, so the declined cases carry roughly three times the error density of the retained ones.

The escalated cases share a diagnosis. Nine of the eleven had no retrievable final regulatory text, only a notification summary and a link resolving to a national portal index; the other two carried a corrigendum whose relationship to the base measure could not be determined from the circulated symbols. In each, the correct answer is to contact the enquiry point, which is what the escalation item instructs.

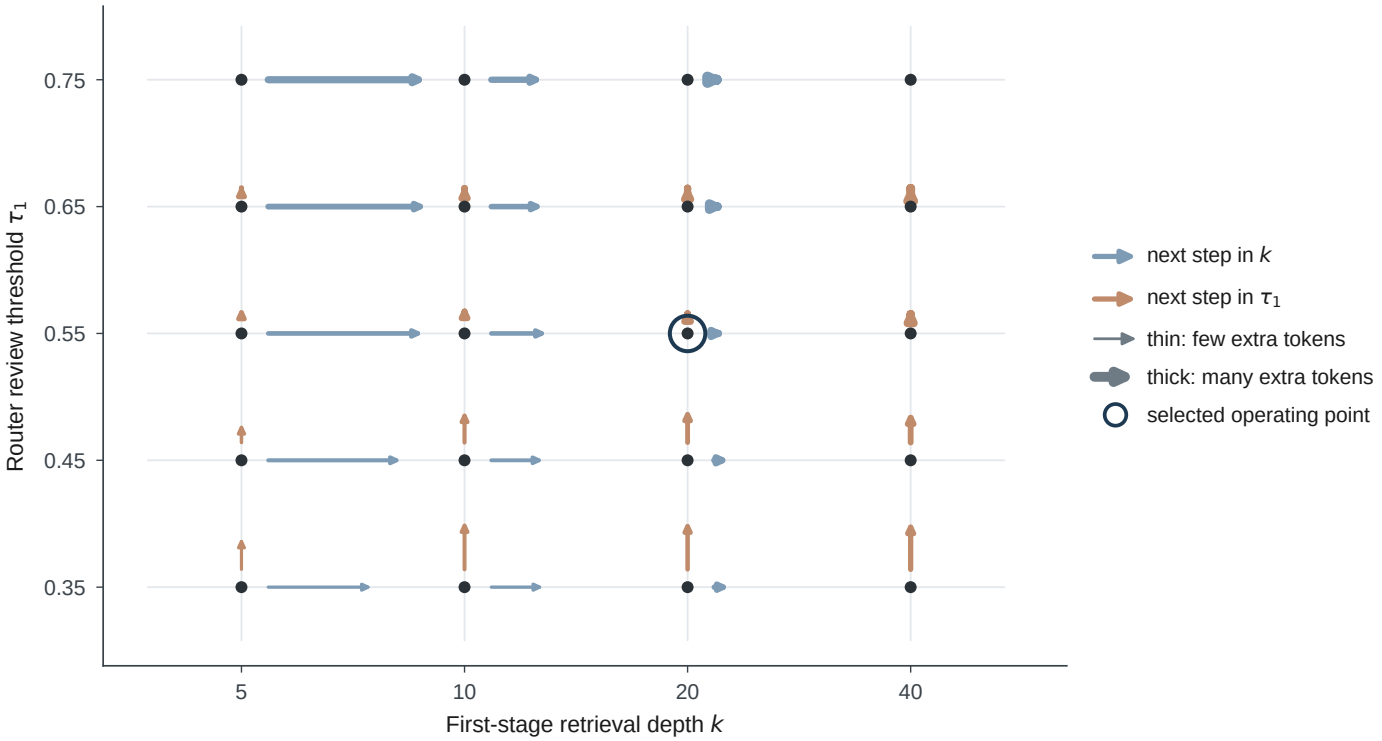


Figure 10. Return field over the router review threshold τ_1, and the first-stage retrieval depth k. The evidence-gate threshold θ is a separate control and is swept in Table 8. At each of the 20 configurations, two arrows show what the next step of each control delivers: arrow length is the citation support gained in percentage points, and arrow thickness is the additional tokens, on the scales given in the key. Arrows shorten and thicken towards the upper right, and the ring marks the selected operating point.

5.8. Expert Assessment

Would a trade compliance professional prefer these reports? Table 11 records the paired assessment on 72 cases. GRACE scores 4.16 of 5 overall against 3.94 for the always-on pipeline and 3.60 for the domain-adapted single agent, and the margin of 0.56 points over the latter is significant with an effect size of 0.58; the paired difference intervals are $[0.42, 0.70]$ against the single agent, with Holm-adjusted $p < 0.001$, and $[0.07, 0.36]$ against the always-on pipeline, with adjusted $p = 0.008$ and Cliff's $\delta = 0.31$. The widest gap, 0.91 points, is in uncertainty appropriateness, where the single-agent system was repeatedly marked down for stating obligations categorically when the notified document supplied only a draft. Sustainability relevance shows the narrowest gap, unsurprisingly: all three systems received the same environmental objective metadata.

Inter-assessor agreement was highest on evidence support and regulatory fidelity, at $\kappa = 0.80$ and $\kappa = 0.78$, where the judgement reduces to checking whether a cited passage says what the report claims, and lowest on sustainability relevance at $\kappa = 0.63$.

Table 12 sets the assessed scores beside what each system costs to produce them. Measured against the domain-adapted single agent, GRACE requires 1.46 thousand additional tokens for each additional point of citation support; the always-on pipeline requires 5.21 thousand, and the hierarchical-audit configuration 5.17 thousand, roughly three and a half times as much. Two systems fall off the efficient frontier once cost is counted in tokens, and a third falls off once it is counted in money: the frontier commercial model is cheaper in tokens but 8.8 times dearer per completed case.

Table 11. Expert assessment on 72 paired cases, two independent assessors with a third arbitrating, five-point scale; system identity concealed and presentation order randomised. Values are means with standard deviations; κ is the quadratic-weighted Cohen's κ between the two primary assessors before arbitration; the means are computed after arbitration. The best value in each row is in bold.

Dimension	B2	B3	GRACE	κ
Regulatory fidelity	3.71 ± 0.62	4.02 ± 0.57	**4.19 ± 0.54**	0.78
Evidence support	3.64 ± 0.68	4.11 ± 0.59	**4.31 ± 0.51**	0.80
Requirement completeness	3.58 ± 0.71	3.96 ± 0.63	**4.12 ± 0.58**	0.72
Business actionability	3.49 ± 0.74	3.78 ± 0.69	**4.07 ± 0.61**	0.66
Uncertainty appropriateness	3.33 ± 0.79	3.85 ± 0.66	**4.24 ± 0.55**	0.69
Sustainability relevance	3.82 ± 0.60	3.91 ± 0.58	**4.03 ± 0.56**	0.63
Overall	3.60	3.94	**4.16**	0.71

Table 12. Quality against computational cost. Efficiency is the additional tokens spent for each additional point of citation support over the single-agent baseline, so lower is better. The last column reports whether a system lies on the efficient frontier in tokens and in money; the frontier model survives the first test and fails the second.

System	Citation Supp. (%)	Expert Overall	Tokens (k)	Latency (s)	USD/Case	Tokens per Added Point	On Frontier
B0 Direct LLM	31.6	—	3.1	4.2	0.008	—	tokens, cost
B1 Generic RAG	79.2	—	11.8	9.6	0.026	—	tokens, cost
B2 Domain single-agent	87.4	3.60	14.3	12.4	0.031	reference	tokens, cost
B4 Hierarchical audit	89.8	—	26.7	24.1	0.058	5.17	neither
B3 Fixed multi-agent	91.3	3.94	34.6	31.8	0.074	5.21	neither
UB Frontier + hybrid	88.1	—	16.2	14.9	0.412	2.71	tokens only
GRACE	92.6	4.16	21.9	20.8	0.047	1.46	tokens, cost

5.9. Residual Failures

Where does the remaining error budget go? Table 13 classifies failures across the 214 families. Action-plan omission remains the most frequent category for every system, at 27 cases for GRACE against 47 for the single agent, with a consistent cause: requirements expressed as cross-references to a national standard that the notified document names but does not reproduce. Unsupported obligations fall from 41 to 19 cases and miscalibrated confidence from 39 to 14, the two categories the evidence gate and escalation module were built to address.

Table 13. Residual failure incidence over the 214 evaluation families. Counts are the number of cases exhibiting each category and do not sum to the case total because a case may exhibit more than one. The hand-off row is the only category where the multi-agent configurations carry a higher incidence.

Failure Category	B2	B3	GRACE
Measure-card parsing	18	13	12
Evidence retrieval miss	33	24	21
Superseded-version use	17	9	8
Unsupported obligation	41	22	19
Hand-off or schema fault	6	14	9
Action-plan omission	47	31	27
Miscalibrated confidence	39	19	14

Hand-off and schema faults are the one category where the multi-agent configuration carries higher incidence, at 9 cases against 6—the structural cost of passing typed

objects between components—and the validator repaired 8 of the 9 within its retry budget. Superseded-version use is nearly eliminated, at 8 cases against 17.

5.10. Case Illustration

One tier-2 case makes the mechanism concrete; Appendix A reproduces it in full. A notified technical regulation on minimum energy performance and comparative labelling for a household appliance was circulated as a base notification, an addendum extending the comment period and a revision altering two thresholds. Preprocessing collapsed the three symbols into one family and identified the revision as admissible; the router scored the case at $u = 0.71$ and sent it to review, and the inspector then removed two requirements, one drawn from the superseded base notification specifying a threshold the revision had changed, and one resting only on a committee intervention, which was rewritten as a question for the enquiry point. The final report carried nine requirements, 17 actions with owners and dates, three information gaps and two escalation items; the single-agent system retained both removed requirements and cited the superseded threshold as binding. The difference is not one of fluency—both reports read well, and one would have sent a product to test against a threshold that no longer applies.

6. Discussion

The results speak to a specific question in sustainable business practice: whether the transparency infrastructure built around the TBT Agreement can be converted into operational capability at a cost that firms without regulatory affairs departments can bear. The evidence here suggests it can at the level of task performance, and locates the value precisely; whether that hardens into durable organisational capability or environmental performance is a question for longitudinal deployment. Publishing notifications is necessary and does not by itself close the gap, which is consistent with UNCTAD's finding that better regulatory transparency would reduce non-tariff-measure trade costs by roughly 19% rather than eliminating them [3]. The residual is interpretive work, and interpretive work is what the system industrialises. At USD 0.047 per completed case, under the reported hardware utilisation and pricing assumptions, a trade association can screen the several thousand notifications relevant to its membership each year for less than the cost of a single consulting engagement, and route only the 5.1% of cases the router escalates to professional review.

Attribution is bounded to the evaluated stack and separates three sources. Document structuring decides whether the right record is reached, with its two mechanisms acting on different failure modes: retriever adaptation moves recall by 8.4 points, and family normalisation moves current-version retrieval by 14.2. Orchestration decides whether retrieved material becomes a defensible obligation, authority weighting and the gate carrying the largest movements in support. Routing primarily controls computational allocation; relative to the always-on B3 pipeline, it also produces small quality gains, consistent with avoiding unnecessary review on easier cases. One-component ablations report sensitivity and need not be additive, and the authority weights, $\theta = 0.62$ and the router thresholds are development-selected operating points: what is claimed is the principle, not these values.

Table 14 puts that separation in one place, taking each entry from the one-at-a-time ablation of Table 7 against the full system. Three readings follow. The layer B2 already carries is what separates a generic pipeline from a competent one, and it is the layer that moves coverage as well as support. The layer specific to GRACE buys reliability, with the gate being the largest single contributor to both support and the unsupported-claim rate and the only mechanism that trades coverage for it. Routing returns the cost of those checks

rather than adding quality, which is why the net token increase over B2 is 7.6 thousand rather than the 20.3 thousand an always-on pipeline would spend.

Table 14. Contribution of each mechanism at the selected operating point, separated by whether the single-agent baseline B2 also carries it. Entries are the full system minus the configuration with that mechanism removed (Table 7), so a positive value on citation support or action coverage and a negative value on unsupported claims are improvements, and a negative token entry is a saving. They are one-at-a-time sensitivities and are not additive: the final row is the direct GRACE-over-B2 difference of Tables 4 and 5. Absolute levels also depend on the fixed model stack and the development-selected thresholds; freezing the generation backbone lowers both architectures by two to three points while leaving the margin at 5.2 points (Table A3).

Mechanism	In B2	Δ CSR	Δ RAC	Δ Unsupp.	Δ Tokens (k)
Domain adaptation: shared with the single-agent baseline					
Retriever domain adaptation	yes	+5.7	+5.3	−3.2	−0.5
Notification-family and date normalisation	yes	+4.3	+2.6	−2.4	+1.1
Source-authority weighting	yes	+5.4	+0.8	−3.7	+0.4
Evidence grounding and orchestration: specific to GRACE					
Deterministic evidence gate	no	+8.0	−1.2	−7.1	+2.3
Escalation module	no	+0.8	+0.4	−1.5	+0.5
Adaptive router	no	+1.3	+1.3	−0.7	−12.7
Net GRACE over B2	—	+5.2	+5.5	−3.9	+7.6

Three findings carry beyond this application. The first concerns what verification is worth. Removing the evidence gate raised apparent requirement coverage by 1.2 points while citation support collapsed by 8.0 points and unsupported claims more than doubled. Any evaluation of a compliance assistant that reports coverage without reporting grounding will therefore reward the wrong system, and the bias runs towards assistants that assert more and support less. This generalises the concern raised by process-oriented evaluation of agentic systems [33] to a setting where the cost of an unsupported assertion is a laboratory invoice or a rejected consignment.

The second concerns where structure beats capability. A frozen frontier model with the same hybrid retrieval matched the domain-adapted single agent on requirement coverage, and fell 2.5 points behind it on current-version retrieval. That comparison is descriptive rather than controlled, since the commercial model's training is not observable: it shows that scale alone did not remove version-handling errors here, not that domain adaptation generally beats larger models. For practitioners choosing between commissioning domain adaptation and waiting for a stronger general model, the distinction is actionable: tasks defined by document conventions rather than by language understanding may not be resolved by model scaling alone.

The third concerns the shape of the efficiency argument in multi-agent systems. Conditional activation cut tokens by 36.7% and latency by 34.6% relative to the always-on pipeline while improving citation support, directionally consistent with the 29.68% token reduction reported for runtime supervision on general agentic benchmarks [36]. What the tier breakdown adds is that the saving is not uniform and should not be reported as if it were. On easy cases, the reduction reached 52%; on the hardest tier, it was 17% because those cases genuinely need the inspector. An aggregate efficiency figure conceals whether a router is discriminating or merely skipping work.

For a compliance function, the practical question is where the system sits in an existing workflow, and the routing distribution answers it. Roughly 6 cases in 10 complete on the fast path, the measures an officer would otherwise triage by hand before setting aside. Four in ten receive automated review, with each obligation bound to a paragraph, so the task

shifts from finding the requirement to checking the binding. A total of 1 in 20 is escalated with a specific question attached, a volume a small firm can afford to send to counsel or a testing body. Ownership of the checklist must stay with the human who signs it.

Traceability is best read as a repeatable managerial routine rather than a property of one report: versioned evidence links and uncertainty records strengthen sensing, owner- and deadline-bound actions support seizing by attaching an accountable function to each decision, and the resulting revisions to specifications, supplier declarations and technical files are where adaptation becomes visible in the product. What a manager can monitor is time to triage, the share of actions carrying tier-A to tier-C evidence, blocking questions unresolved at design freeze, missed deadlines and rework caused by version changes. The offline study establishes these antecedents, not eventual environmental performance.

There is also a narrow message for the design of transparency instruments. The mechanisms producing the largest gains all compensate for structure that exists in the records but is not exposed in machine-readable form: family normalisation reconstructs, from string patterns in document symbols, a supersession relationship the notifying member already knows; the date triple reconstructs, from free text, a distinction the member also knows; scope reconciliation compensates for classification codes absent in 31.3% of the families examined. Each is a place where a firm without engineering capacity absorbs a cost that a structured field would remove.

Several limitations bound these conclusions. The system produces no assessment of whether any measure is consistent with WTO obligations, and its output is not legal advice. Notifications are not enacted law, and a proposed entry-into-force date may move, so a report is accurate as of its evidence retrieval date and no longer. The relative comparisons hold within one stack of generation, embedding and reranking models and one set of development-tuned thresholds, so absolute levels may differ under another stack. Translation-induced error was not measured separately, so performance on families whose implementing text exists only in a language absent from the embedding model's 34 is not established. The evaluation covers three low-sensitivity regulatory families and excludes military, dual-use, semiconductor, critical-minerals and export-control domains by design, and its 214 families over-represent complex cases, so the aggregate is a challenge-set rather than a deployment estimate. The corpus reflects what members chose to notify, so measures never notified are invisible by construction. Sensing, evidence support and action planning are evaluated offline, which establishes neither organisational adoption nor export or environmental outcomes. Reproduction is constrained by the closed adaptation strategy: sources, audit counts, architecture, schemas, metrics and aggregate results are reported in full, as are the split manifest, pair-construction rules, representative prompts and orchestration pseudocode of Appendix B; adapter weights and raw traces are not, so the material supports procedural replication rather than bit-for-bit reproduction of the reported checkpoints.

Two threats to validity deserve statement. The reference checklists encode the research team's view of completeness, which the expert assessment was included partly to test; the two rankings agree. With two primary assessors and 72 cases, the comparison has adequate power for the 0.56-point gap against the single-agent system and limited power for the 0.22-point gap against the always-on pipeline, and it speaks to report quality rather than to adoption preferences across user groups.

7. Conclusions

This study built and evaluated a WTO-informed multi-agent decision-support system that converts cross-border green product regulations into evidence-grounded, human-reviewable compliance actions for firms in emerging markets. On 214 held-out notification

families spanning energy efficiency, restricted substances and packaging circularity, it grounded 92.6% of its requirement statements in admissible official sources against 87.4% for a single-agent system carrying identical domain adaptation, cut unsupported assertions by 45%, retrieved the currently admissible document in 95.8% of cases, and consumed 36.7% fewer tokens than an always-on pipeline of the same agents. Blind trade-law assessors rated its reports 4.16 of 5 against 3.60, with the largest margin on how the system handles what it does not know. Controlled baselines and ablations associate the within-stack differences with identifiable mechanisms, subject to interaction among them, and they fall into two classes that this evaluation keeps apart. Domain adaptation—family normalisation and contrastive retriever adaptation—separates a generic retrieval pipeline from a competent one, carrying B1 to B2 and raising current-version retrieval by 11.2 points and citation support by 8.2. The single-agent baseline already carries both, so the margin over it rests on the orchestration layer, where conditional verification, the gate and escalation add 5.2 points of citation support and 5.5 points of requirement-action coverage, delivered by the router at 36.7% fewer tokens than running the same components unconditionally.

Three extensions follow. Cross-lingual evidence handling is the most consequential, since much implementing text exists only in national languages. Longitudinal deployment would test whether gains in action completeness translate into fewer missed deadlines and redundant tests, which no offline metric observes. Extending the authority model to conformity assessment recognition arrangements would address the largest residual failure category. Transparency instruments deliver their value at the moment a document becomes an action, and the evaluated architecture makes that conversion more reliable and traceable at the reported cost, though its organisational consequences remain to be tested in deployment.

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Informed Consent Statement: Not applicable.

Data Availability Statement: This study analyses existing public records made available by the World Trade Organization through the ePing SPS & TBT Platform and the Environmental Database, both cited in the reference list. The split manifest, representative task prompts, orchestration pseudocode and fitted router coefficients are reported in Appendix B. Codes are available from the corresponding author on reasonable request.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CSR	Citation support rate
ESAC	Evidence-supported action coverage
NLI	Natural language inference
RAC	Requirement-action coverage
RAG	Retrieval-augmented generation
RR	Requirement retention

RRF	Reciprocal rank fusion
SPS	Sanitary and Phytosanitary
TBT	Technical Barriers to Trade

Appendix A. Worked Compliance Reports

This appendix reproduces system output rather than describing it: three cases, one per regulatory family, in the form a compliance officer receives them. Two redactions apply, and nothing else is altered—the notifying market becomes [MKT], and product descriptors are generalised to a category—while document symbols retain their real structure and evidence identifiers carry the authority tier in brackets.

Appendix A.1. Case EE-07: Energy Performance and Comparative Labelling

The input is a notified technical regulation setting minimum energy performance thresholds and a mandatory comparative label for a household laundry appliance category, circulated as a base notification, an addendum extending the comment period and a revision altering two thresholds; the firm profile describes three product variants and a launch two weeks after entry into force. Listing A1 is the measure card emitted after preprocessing resolved the family.

Listing A1. Measure card emitted for case EE-07, reproduced verbatim.

```
{
 ‘‘document_family’’:    ‘‘G/TBT/N/[MKT]/412’’,
 ‘‘admissible_document’’:  ‘‘G/TBT/N/[MKT]/412/Rev.1’’,
 ‘‘version_status’’:     ‘‘revision’’,
 ‘‘family_members’’:     [‘‘G/TBT/N/[MKT]/412’’,
                  ‘‘G/TBT/N/[MKT]/412/Add.1’’,
                  ‘‘G/TBT/N/[MKT]/412/Rev.1’’],
 ‘‘market’’:           ‘‘[MKT]’’,
 ‘‘instrument_type’’:    ‘‘technical_regulation’’,
 ‘‘regulatory_family’’:  ‘‘energy_efficiency’’,
 ‘‘product_scope’’:     [‘‘household laundry appliances, rated
                  capacity 2.0-9.0 kg’’, ‘‘excludes
                  commercial machines’’],
 ‘‘hs_codes’’:         [‘‘8450.11’’, ‘‘8450.12’’, ‘‘8450.19’’],
 ‘‘requirement_types’’:  [‘‘testing’’, ‘‘labelling’’,
                  ‘‘documentation’’, ‘‘registration’’],
 ‘‘key_dates’’:  {
  ‘‘comment_deadline’’:   ‘‘2025-11-14’’,
  ‘‘proposed_adoption’’:  ‘‘2026-03-02’’,
  ‘‘adopted’’:          ‘‘2026-03-30’’,
  ‘‘entry_into_force’’:   ‘‘2026-07-13’’,
  ‘‘label_format_mandatory’’:  null
 },
 ‘‘uncertain_fields’’:   [‘‘key_dates.label_format_mandatory’’,
                  ‘‘product_scope.capacity_below_2.0kg’’]
}
```

The router scored the case at $u = 0.71$ and sent it to the review path. Table A1 gives five of the nine requirements that survived the gate, each with the span the retriever bound to it.

Table A1. Case EE-07, five of nine retained requirements as emitted, with the bound evidence span. Tier B denotes the notified regulatory document, and tier C denotes an official implementation note. Windows are weeks relative to entry into force, and action numbers follow the order in which the planner emitted the checklist.

#	Requirement as Emitted	Evidence	Bound Span	Action, Window
R1	Energy efficiency index at or below the value for the applicable rated capacity class	Rev.1, Annex II tbl.2 [B]	"…the energy efficiency index shall not exceed the value in Table 2 of Annex II for the applicable rated capacity class"	A4, −27 to −21
R2	Declared values verified by a body designated under Article 12	Rev.1, Art. 12(1) [B]	"…the declared values shall be verified by a conformity assessment body designated under Article 12"	A3, −30 to −27
R3	Comparative label in the Annex IV format, including in distance selling	Rev.1, Annex IV [B]	"…the label shall conform to the format in Annex IV and be displayed in any visual advertisement and distance-selling offer"	A6, −20 to −14
R4	Technical documentation retained 10 years after the last unit is placed on the market	Impl. note §4.3 [C]	"…technical documentation shall be kept for inspection for 10 years after the last unit was placed on the market"	A8, −16 to −6
R5	Model entered in the product database before placing on the market	Rev.1, Art. 9(2) [B]	"…before placing a model on the market the supplier shall enter the information in Annex VI in the product database"	A11, −6 to −3

Two draft requirements did not survive. The first read "Standby power shall not exceed 0.50 W in off-mode"; its only support was Annex II of the base notification, whose threshold the revision replaced. The second read "Test reports must be issued by a laboratory accredited in [MKT]"; the only retrieved support was a Committee intervention, which is tier D and cannot establish an obligation. Listing A2 reproduces what the system returned in their place. The single-agent system retained both and printed the second without qualification, together with the instruction "Complete accredited testing in [MKT] by 2026-05-04", for which no retrieved passage supplies a date.

Listing A2. Information gaps and escalation questions returned for case EE-07, reproduced verbatim.

```
INFORMATION GAPS
 G1  key_dates.label_format_mandatory is stated in neither
     G/TBT/N/[MKT]/412/Rev.1 nor the implementation note.
     Impact:  A6 has no fixed deadline; scheduled against
     entry into force as a conservative proxy.
 G2  Scope for rated capacities below 2.0 kg is not addressed.
     Impact:  one of three declared variants may fall outside.
ESCALATION QUESTIONS
 E1  To:  enquiry point of [MKT].
     ‘‘Does [MKT] accept test reports issued by a body
       accredited under a mutual recognition arrangement, or
       must the body be designated under Article 12 of the
       notified measure?’’
     blocks:  A3, A4, A10
     raised:  sole retrieved support is tier D and cannot
          establish an obligation
 E2  To:  counsel.
     ‘‘Does Article 14 permit units manufactured before
       13 July 2026 to be sold through after that date?’’
     blocks:  launch date confirmation
```

Appendix A.2. Case RS-22: Restricted Substances in a Consumer Electronics Accessory

The notification exists as a single version with a complete notified document, and the router assigned the case to the fast path at $u = 0.34$. The substantive point is visible in one requirement entry, reproduced as Listing A3: the restriction attaches to homogeneous

materials, not to the finished article, and the planner acted on that distinction rather than scheduling a single product test.

Listing A3. Requirement entry and escalation question for case RS-22, reproduced verbatim.

```
REQUIREMENT R3
 statement  :  Each homogeneous material in the accessory shall
          contain no more than the concentration limits set
          out in Annex I, by weight.
 evidence   :  G/TBT/N/[MKT]/389 Art.4(1), Annex I  [tier B]
          ‘‘...concentration values by weight in homogeneous
           materials shall not exceed the limits in Annex I...’’
 scope_note :  applies at material level, not at article level
 actions    :  A1 decompose declared bill of materials into
               homogeneous materials [product engineering]
          A3 test each distinct material, not the assembly
              [quality assurance]
 confidence :  high    human confirmation required:  no
ESCALATION QUESTION
 E1  To:  enquiry point of [MKT].
      ‘‘Does [MKT] accept analytical reports prepared under
       the international test method referenced in Article
       4(3) when the testing body is located outside [MKT]?’’
      blocks:  A3 scheduling
```

Appendix A.3. Case PC-14: Packaging Marking and Recyclability Declaration

The notified document is a four-page summary whose operative criteria sit in a national instrument it cites without reproducing, which placed the case in tier 3. Listing A4 shows how the system handled the hole: it withheld the requirement depending on the missing instrument, named that instrument in the escalation, and listed the actions still specifiable without it. The single-agent baseline instead produced a recyclability assessment action citing generic criteria the notification never referenced.

Listing A4. Blocking information gap returned for case PC-14, reproduced verbatim.

```
INFORMATION GAP G1  [blocking]
 requirement      :  R2 ‘‘packaging shall be assessed as
                 recyclable according to the national
                 criteria’’
 cited_instrument   :  G/TBT/N/[MKT]/451 refers to
                 ‘‘Resolution No.  118, Annex III’’ and does
                 not reproduce it
 retrieved_candidates:  2, both tier D (Committee minutes)
 status         :  withheld from the requirement map
 escalated_as     :  E1, to the enquiry point of [MKT],
                 requesting the applicable version of
                 Resolution No.  118
 unaffected_actions :  A1, A4, A5, A6
```

Read together, the three outputs show the property the metrics summarise. Each obligation carries the span that supports it, each date is labelled by its legal status, and each thing the system could not settle is addressed to somebody who can settle it.

Appendix B. Reproducibility Protocol and Supplementary Results

Appendix B.1. Corpus Splits and Training Pairs

Splitting is by notification family throughout, so no revision of a family used for adaptation reappears at evaluation time. Table A2 states the manifest: the 5424 families of the sampling pool partition exactly into three splits, and the 66 reserved 2016–2023 families are withdrawn before the training split is formed. Training positives come from three relations—title and product description paired with the notified document, requirement

summary paired with its supporting passage, and base notification paired with a family member—in the proportion 15,204:13,890:9648, each drawing seven hard negatives under the rule of Section 3.

Table A2. Split manifest. Family counts partition the 5424-family sampling pool of Section 4.1. Pairs and instances are counted after deduplication; the held-out split contributes no training material and informs no threshold, checkpoint, prompt or router decision.

Split	Families	Positives	Hard Negatives	Generation Instances	Hand Reviewed	Circulation Years
Training	4004	38,742	271,194	17,506	312	2009–2023
Development	1206	9864	69,048	4318	0	2024
Held-out evaluation	214	0	0	0	0	2016–2023 (66) and 2025–2026 (148)

Appendix B.2. Prompts and Orchestration

Listing A5 gives the four task prompts in representative form together with the orchestration procedure, its routing thresholds, retry budget, stopping rule and the fitted router coefficients. The prompts carry the field constraints, evidence conditions and output schema of those executed, and they differ only in wording and in the omission of internal identifier formats.

Listing A5. Representative task prompts and the orchestration procedure.

```
INTERPRETER. Populate a field only from supplied evidence and return its
 evidence ids.  Never infer missing scope, dates, status or conformity
 requirements:  use null and list the field in uncertain_fields.
 IN  reference_date, notification_family, admissible_document,
     firm_profile, evidence_passages(passage_id, authority_tier, text)
 OUT JSON conforming exactly to schema_measure_card_v3.json
INSPECTOR. Verify per requirement:  (1) a tier A-C passage entails it;
 (2) no superseded family member carries operative support; (3) each
 date is labelled proposed/adopted/in force and matches its source;
 (4) no committee discussion or trade concern is stated as an
 obligation.  Return KEEP or WITHHOLD, evidence ids, score, reason
 code; never rewrite a WITHHOLD as an obligation.
PLANNER. Map only KEEP requirements to actions carrying action_text,
 accountable_function, time_window, prerequisite, requirement_id and
 inherited evidence ids.  A missing fact becomes a question, never an
 instruction.
REPAIR. Repair syntax and schema violations only; never add, remove or
 reinterpret obligations or evidence bindings.

preprocess:  family <- normalise(symbols); d* <- admissible(family, t);
        dates <- triple(text); scope <- reconcile(hs, ics, text, profile)
retrieve :  C <- rerank(RRF(BM25, dense), k = 20) -> top 8, scored
        by Equations (2) and (3)
interpret :  card <- interpreter(d*, scope, C)
route    :  u <- sigmoid(w'phi + b) by Equation (5); u < 0.55 -> fast:  gate,
        plan;
        0.55 <= u < 0.86 -> review:  inspector, gate, plan;
        u >= 0.86 -> escalate:  inspector, gate, plan, human queue
gate     :  keep(q) by Equation (4); withheld -> gap and escalation question
validate :  schema check; one repair attempt; unrecovered -> incomplete
stop     :  one invocation per agent per case; at most one repair
        attempt per schema failure; no fallback to an ungrounded
        answer

router w, on development-standardised features except the two
 indicators:  top-1 reranker score −0.71; first-to-second margin
 -0.48; evidence coverage ratio −1.34; distinct family documents
 +0.19; missing date components +0.42; scope-ambiguity indicator
 +0.83; unbound draft claims +0.96; schema-validation failure
 +0.57; intercept b = −1.71
```

Appendix B.3. Seed Dispersion and Sensitivity Checks

Dispersion across the five seeds behind each mean in Table 4 is 0.4 to 0.7 percentage points for GRACE, 0.5 to 1.1 for the adapted baselines and 1.7 to 2.6 for direct prompting; it is smaller than the family-level intervals throughout and orders the systems identically. Table A3 tests whether the architectural margin depends on a particular prompt or on the generation adapter. Absolute levels move by at most 0.7 points, the GRACE-over-B2 margin in citation support stays between 5.2 and 5.4, and the system-by-prompt interaction, tested as the system-by-prompt term of a linear mixed model with random intercepts for notification family, is not significant ($p = 0.41$). Freezing the backbone lowers both architectures by two to three points while leaving that margin at 5.2, so the architectural contribution is not an artefact of generation adaptation, although absolute performance bundles the two.

Table A3. Prompt and adaptation sensitivity on the 214 evaluation families and five seeds, with retrieval, schemas and decoding fixed. The P1 row is the as-executed configuration and therefore reproduces the corresponding entries of Table 4. P2 retains only field constraints, and P3 reverses instruction order; the last row removes the QLoRA adapter and changes nothing else.

Condition	Citation Supp.		Action Cov.		Unsupp. Claims	
	B2	GRACE	B2	GRACE	B2	GRACE
P1 as executed	87.4	92.6	82.9	88.4	8.6	4.7
P2 constraints only	86.8	92.1	82.4	88.0	8.9	5.0
P3 reordered	86.5	91.9	82.1	87.8	9.2	5.2
Frozen backbone, P1	84.9	90.1	79.8	85.1	10.8	6.1

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